

Network Security Project

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Introduction

This project focuses on the design, configuration, and verification of a secure enterprise network using Cisco routing, switching, firewall, and server technologies. The network was built to support multiple internal and external segments, including branch networks, guest access, a demilitarized zone (DMZ), and Internet connectivity, while enforcing strong security and access-control policies.

The work begins with assigning IP addresses to all devices to ensure proper Layer 3 connectivity, followed by enabling secure remote management using SSH. Dynamic routing is implemented using OSPF, along with cryptographic authentication to protect routing updates. Switch security features are applied to enhance access-layer protection, and a site-to-site VPN is configured to securely interconnect remote networks.

Firewall technologies are introduced through both router-based security mechanisms and a Cisco ASA firewall. These configurations include zone-based firewall policies, stateless access control lists, Modular Policy Framework (MPF) inspection, Network Address Translation (NAT), and DMZ protection. Server-based authentication is implemented using AAA with RADIUS, supported by a local backup database to ensure availability.

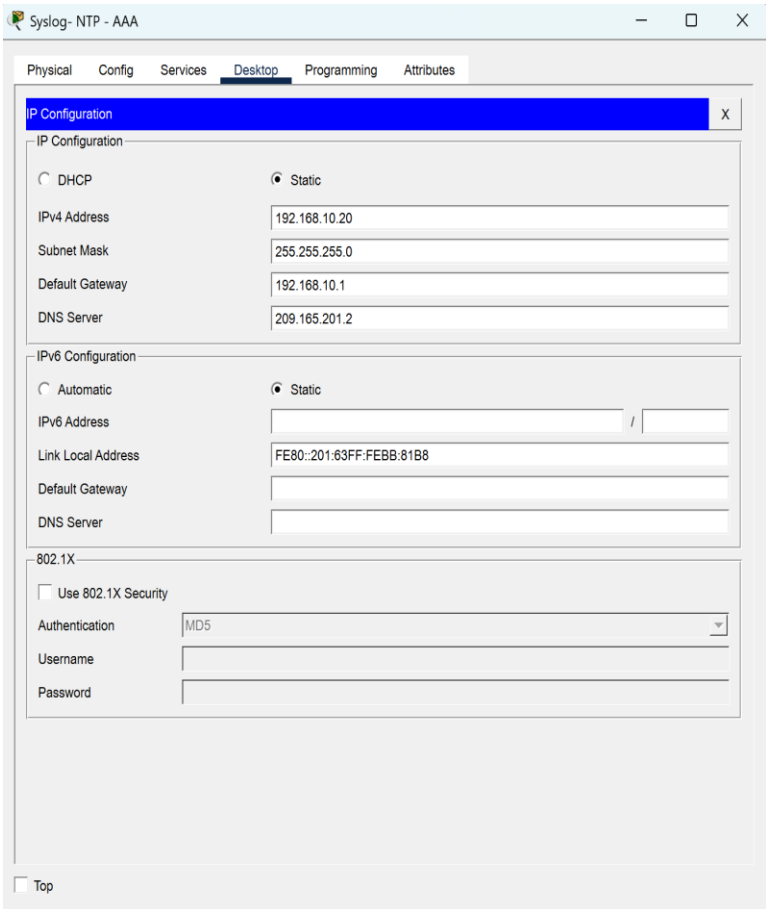
In addition, multiple network services are configured, including FTP, email, web hosting, DNS, DHCP, Syslog, and time management. Logging and timestamp synchronization are used to provide visibility into network events and improve monitoring accuracy. Together, these configurations demonstrate how different networking and security components operate as a unified system to deliver a secure, scalable, and manageable network infrastructure.

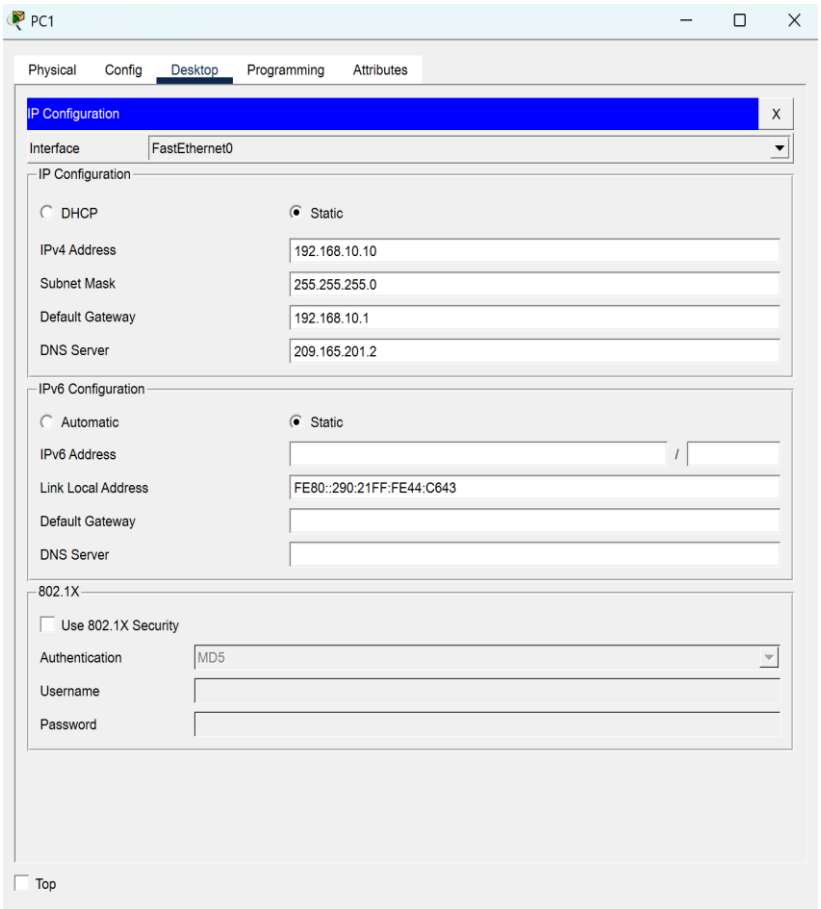
Part 1: IP Address Configuration

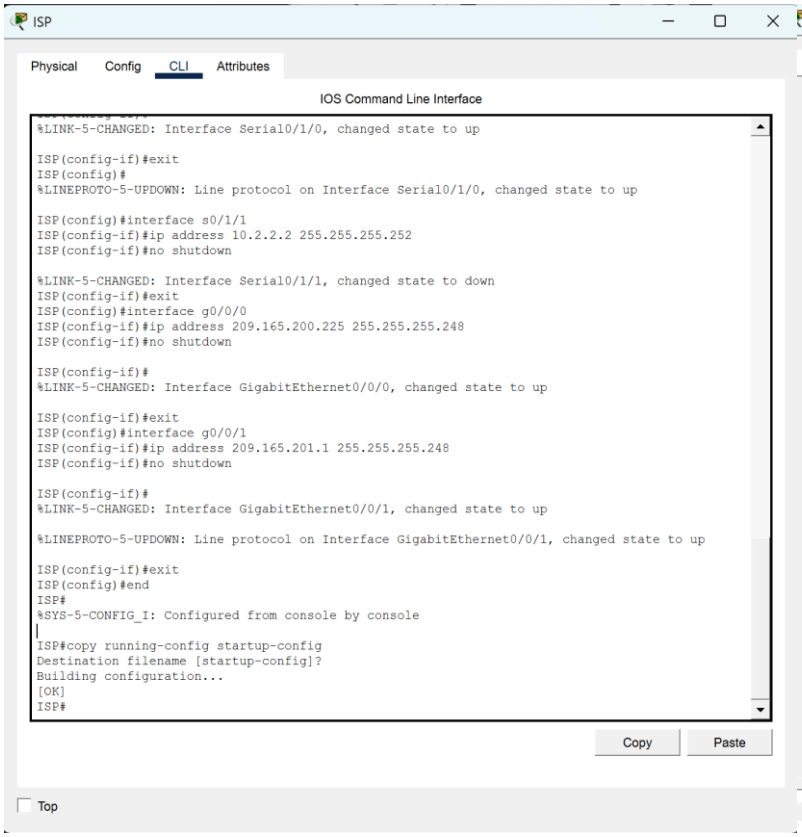
In this part, IPv4 addresses were configured on all network devices according to the provided address table. All PCs and servers were assigned static IP addresses, subnet masks, default gateways, and DNS server addresses to ensure proper communication within the network.

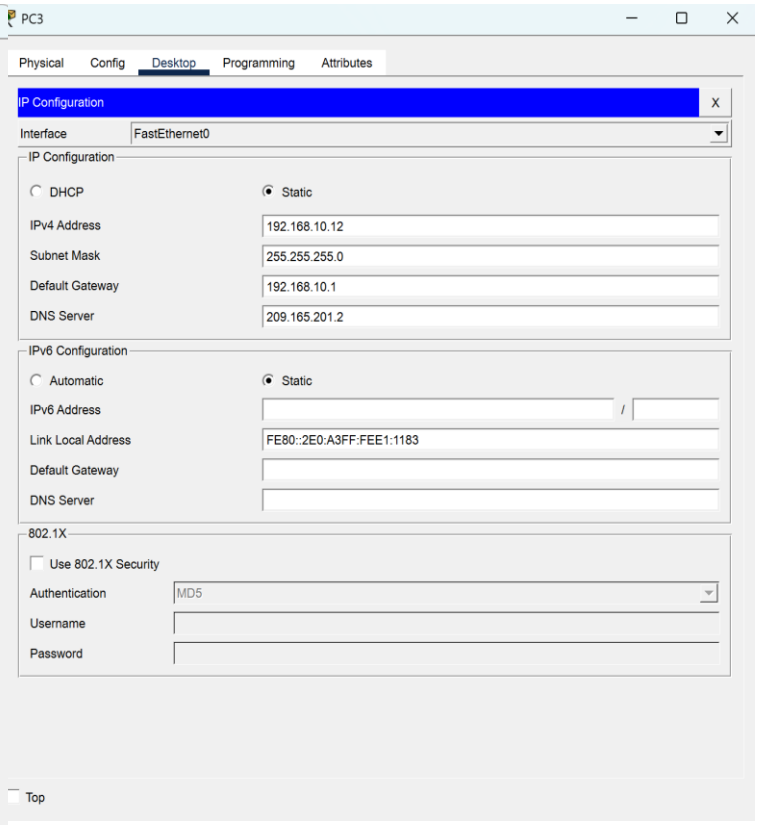
All router interfaces on **R1**, **R2**, and **ISP** were configured with the correct IPv4 addresses for both LAN and WAN connections, and the interfaces were enabled. In addition, the switch **SW1** was configured with a management IP address and a default gateway.

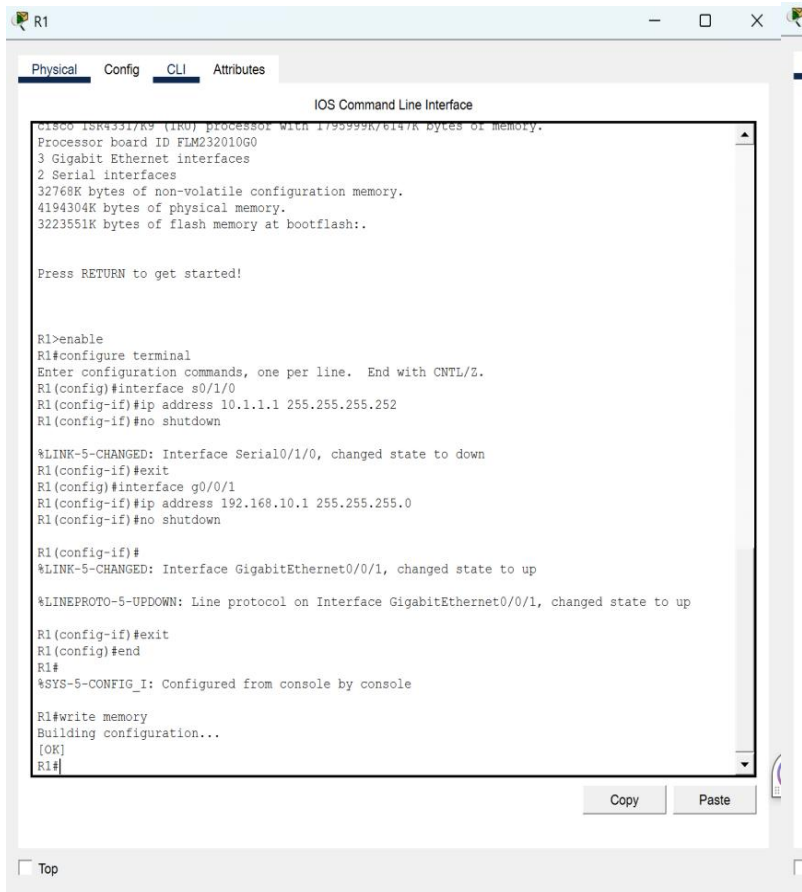
The configuration was verified using the show ip interface brief command on routers and the switch. Connectivity was also tested by successfully pinging the default gateway from a PC, confirming that the IP addressing was correctly implemented and the network was functioning as expected.

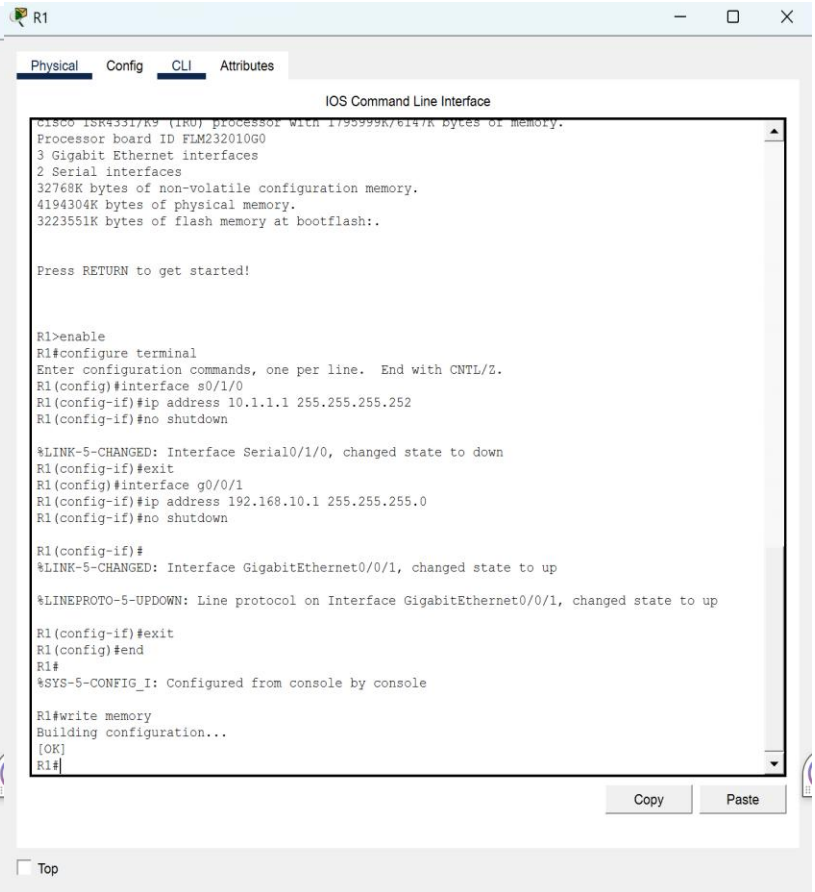


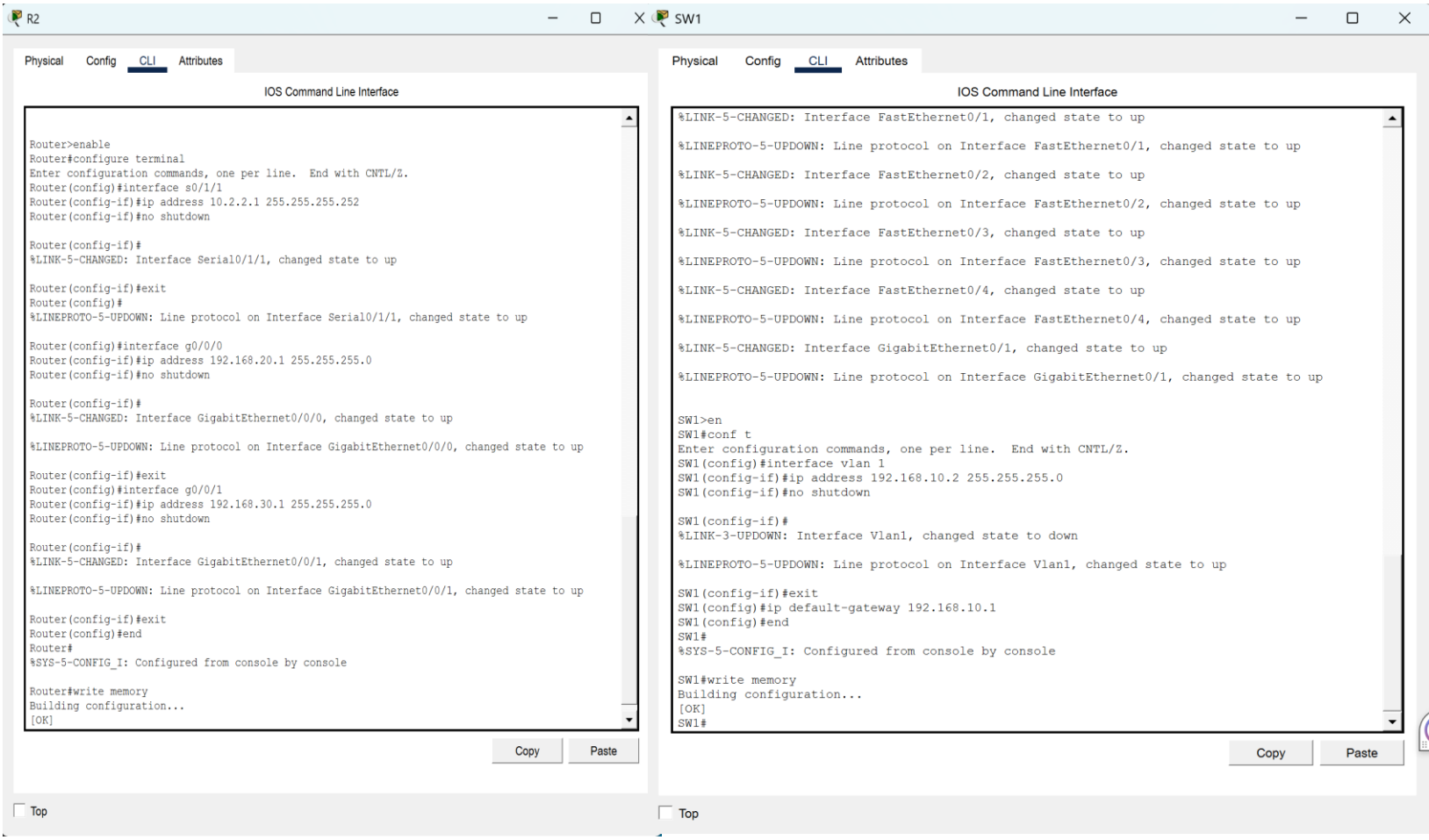



ISP
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
ISP(config-if)#exit
ISP(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
ISP(config)#interface s0/1/1
ISP(config-if)#ip address 10.2.2.2 255.255.255.252
ISP(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
ISP(config-if)#exit
ISP(config)#interface g0/0/0
ISP(config-if)#ip address 209.165.200.225 255.255.255.248
ISP(config-if)#no shutdown
ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
ISP(config-if)#exit
ISP(config)#interface g0/0/1
ISP(config-if)#ip address 209.165.201.1 255.255.255.248
ISP(config-if)#no shutdown
ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
ISP(config-if)#exit
ISP(config)#end
ISP#
%SYS-5-CONFIG_I: Configured from console by console
ISP#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
ISP#
Copy
Paste
Top


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.12
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
209.165.201.2
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:A3FF:FEE1:1183
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Processor board ID FLM232010G0
3 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.
Press RETURN to get started!
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#exit
R1(config)#interface g0/0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
Copy
Paste
Top


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Processor board ID FLM232010G0
3 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.
Press RETURN to get started!
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/1/0
R1(config-if)#ip address 10.1.1.1 255.255.255.252
R1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#exit
R1(config)#interface g0/0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
R1(config-if)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
Copy
Paste
Top

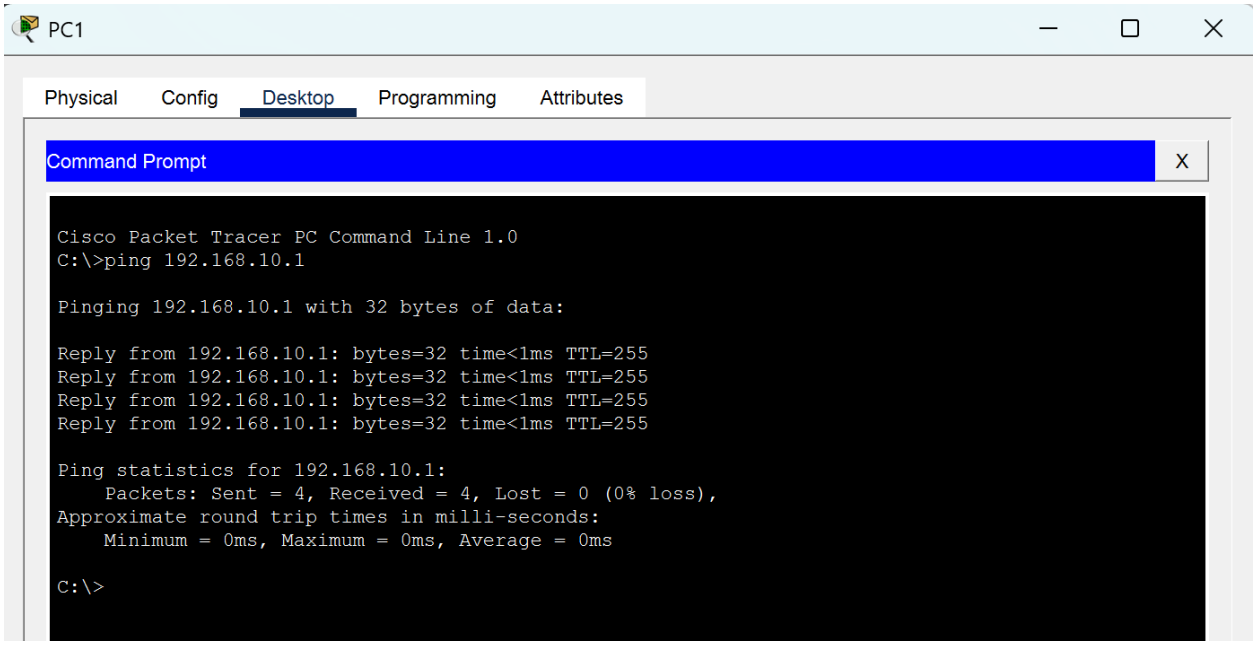



```
R1#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES NVRAM  administratively down down
GigabitEthernet0/0/1   192.168.10.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES NVRAM  administratively down down
Serial0/1/0            10.1.1.1        YES manual administratively down down
Serial0/1/1            10.2.2.2        YES manual administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

```
R2#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.20.1    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES manual up                    up
GigabitEthernet0/0/1.30192.168.30.1    YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            unassigned      YES unset  administratively down down
Serial0/1/1            10.2.2.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
```

```
ISP>en
ISP#show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   209.165.200.225 YES manual up                    up
GigabitEthernet0/0/1   209.165.201.1   YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            10.1.1.2        YES manual down                  down
Serial0/1/1            10.2.2.2        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
ISP#
```

```
SW1>show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual up                    up
FastEthernet0/5        unassigned      YES manual administratively down down
FastEthernet0/6        unassigned      YES manual administratively down down
FastEthernet0/7        unassigned      YES manual administratively down down
FastEthernet0/8        unassigned      YES manual administratively down down
FastEthernet0/9        unassigned      YES manual administratively down down
FastEthernet0/10       unassigned      YES manual administratively down down
FastEthernet0/11       unassigned      YES manual administratively down down
FastEthernet0/12       unassigned      YES manual administratively down down
FastEthernet0/13       unassigned      YES manual administratively down down
FastEthernet0/14       unassigned      YES manual administratively down down
FastEthernet0/15       unassigned      YES manual administratively down down
FastEthernet0/16       unassigned      YES manual administratively down down
FastEthernet0/17       unassigned      YES manual administratively down down
FastEthernet0/18       unassigned      YES manual administratively down down
FastEthernet0/19       unassigned      YES manual administratively down down
FastEthernet0/20       unassigned      YES manual administratively down down
FastEthernet0/21       unassigned      YES manual administratively down down
FastEthernet0/22       unassigned      YES manual administratively down down
FastEthernet0/23       unassigned      YES manual administratively down down
FastEthernet0/24       unassigned      YES manual administratively down down
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual administratively down down
Vlan1                  192.168.10.2    YES manual up                    up
SW1>
```

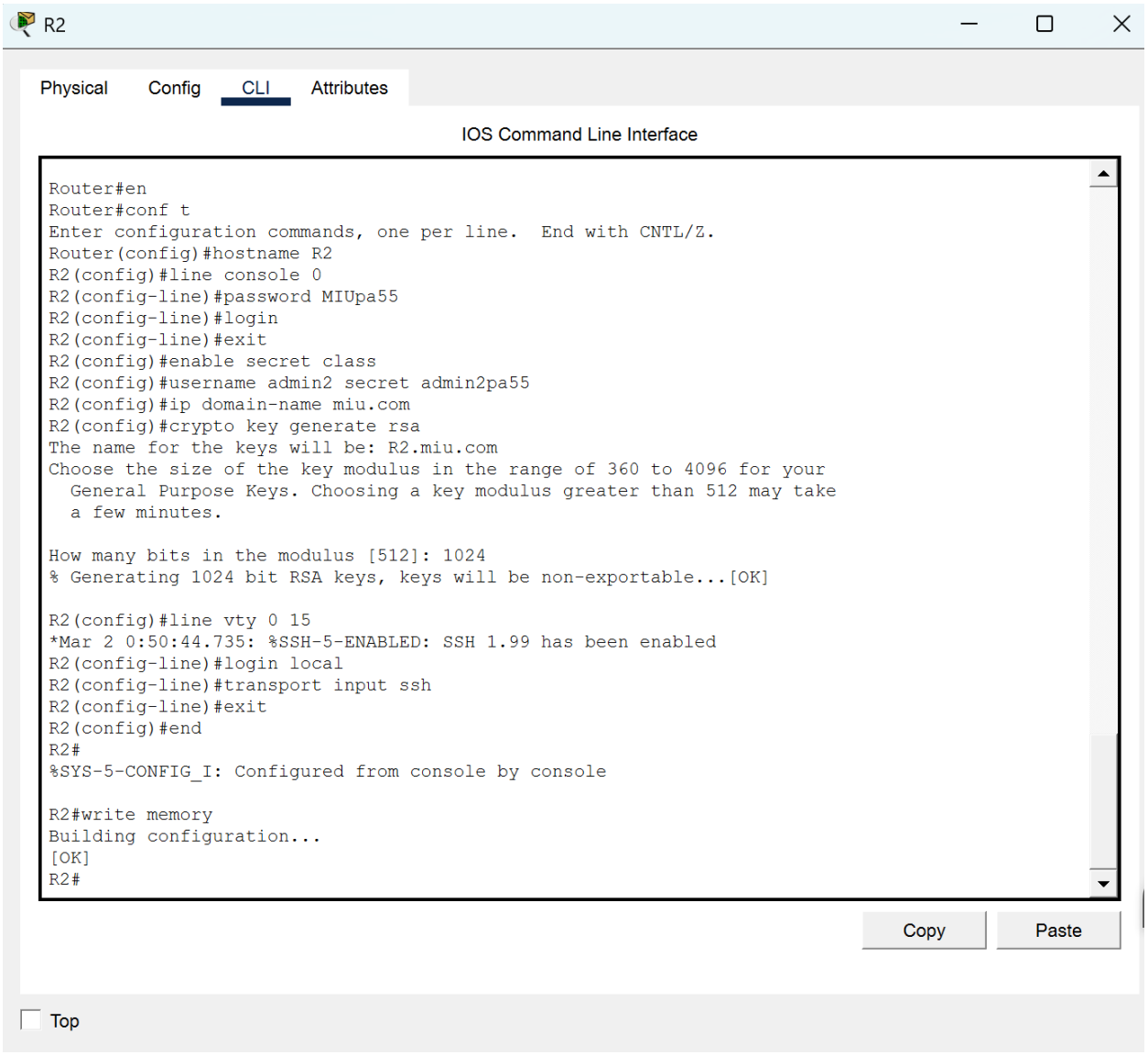


Part 2: Basic Configuration and SSH on R2

Part 2.1: Basic Configuration on R2

In this part of the project, basic configuration was performed on router **R2** to prepare it for secure management. The hostname was configured as **R2** to clearly identify the device within the network topology.

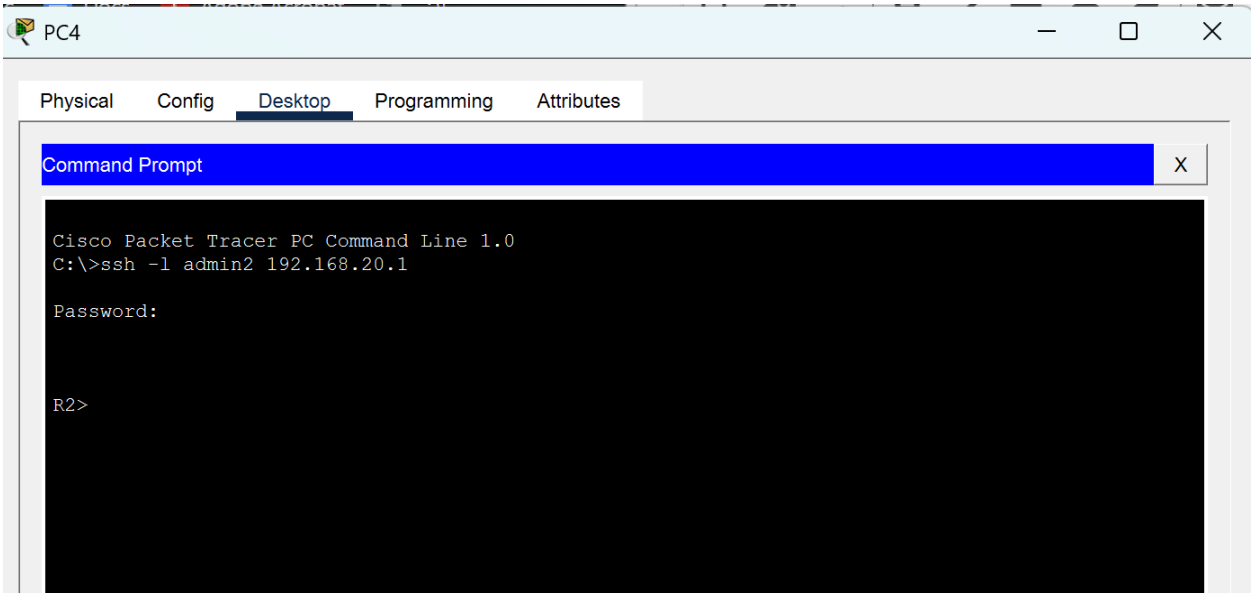
To enhance security, a password was configured on the console line to restrict direct console access. In addition, an encrypted privileged EXEC mode password was set to protect access to configuration mode. These steps ensure that only authorized users can manage the router locally.



Part 2.2: SSH Configuration on R2

Secure Shell (SSH) was configured on R2 to allow encrypted remote access. A local administrative user account was created and stored in the local user database for authentication purposes. The domain name was configured, and RSA encryption keys with a modulus of **1024 bits** were generated to enable SSH functionality.

The VTY lines were then configured to authenticate users using the local database and to accept **only SSH connections**, preventing the use of insecure remote access methods. Finally, SSH access was tested successfully from a PC, confirming that remote management of R2 is secure and functioning correctly.




```
R2>en
Password:
R2#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R2#
```

Part 3: Configure Networks for OSPF Routing

In this part of the project, Open Shortest Path First (OSPF) routing was configured on routers **R1**, **R2**, and **ISP** to enable dynamic routing between all internal networks. OSPF was enabled using process ID **10**, and all relevant networks were advertised in **area 0**.

Each router was configured with appropriate OSPF network statements to ensure that all connected LAN and WAN interfaces participate in the routing process. This allowed routers to automatically exchange routing information and maintain updated routing tables.

In addition, the ISP router was configured with a directly connected default route to the Internet. This default route was then propagated to the internal routers through OSPF updates, allowing internal networks to reach external destinations through the ISP.

After completing the configuration, routing tables were verified to confirm successful OSPF neighbor relationships and correct route propagation across the network.

```
ISP>en
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#router ospf 10
ISP(config-router)#network 10.1.1.0 0.0.0.3 area 0
ISP(config-router)#
01:00:47: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.10.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

ISP(config-router)#network 10.2.2.0 0.0.0.3 area 0
ISP(config-router)#
01:01:08: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.30.1 on Serial0/1/1 from LOADING to FULL,
Loading Done

ISP(config-router)#network 209.165.200.224 0.0.0.7 area 0
ISP(config-router)#network 209.165.201.0 0.0.0.7 area 0
ISP(config-router)#exit
ISP(config)#
```

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 10
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

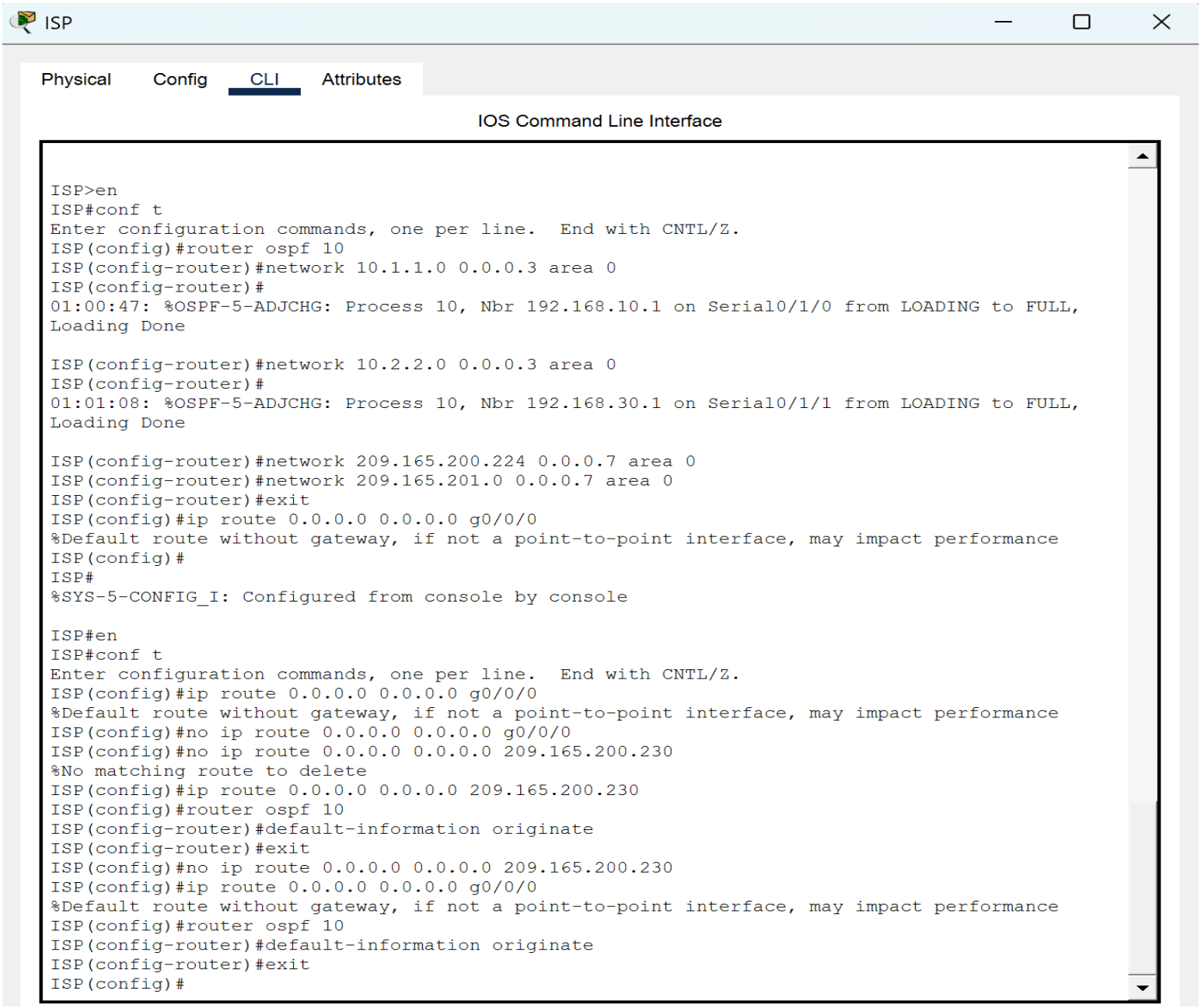
R1(config-router)#network 192.168.10.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#
```

R2

```
R2#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 10
R2(config-router)#network 10.2.2.0 0.0.0.3
% Incomplete command.
R2(config-router)#network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#network 192.168.20.0 0.0.0.255 area 0
R2(config-router)#network 192.168.30.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#
```




```
R1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
209.165.201.1     0   FULL/  -        00:00:39    10.1.1.2        Serial0/1/0
R1#
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.1.1.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.1.1.0/30 is directly connected, Serial0/1/0
L       10.1.1.1/32 is directly connected, Serial0/1/0
O       10.2.2.0/30 [110/128] via 10.1.1.2, 00:04:21, Serial0/1/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/1
O    192.168.20.0/24 [110/129] via 10.1.1.2, 00:04:21, Serial0/1/0
O    192.168.30.0/24 [110/129] via 10.1.1.2, 00:04:21, Serial0/1/0
     209.165.200.0/29 is subnetted, 1 subnets
O       209.165.200.224/29 [110/65] via 10.1.1.2, 00:04:21, Serial0/1/0
     209.165.201.0/29 is subnetted, 1 subnets
O       209.165.201.0/29 [110/65] via 10.1.1.2, 00:04:21, Serial0/1/0
O*E2 0.0.0.0/0 [110/1] via 10.1.1.2, 00:01:11, Serial0/1/0

R1#
```

Part 4: Configure OSPF Routing Protocol Authentication Using SHA256

In this part, OSPF authentication was configured to enhance routing security between all routers. A key chain named **MIU_SEC** with key ID **10** was created on **R1, ISP, and R2**. The authentication key string **MIU_AI_24** was used, and **message-digest authentication** was enabled to protect OSPF updates from unauthorized routers. *Although the task required SHA-256, Cisco Packet Tracer does not fully support SHA authentication for OSPF, so message-digest authentication was used instead*.

After creating the key chain, OSPF authentication was applied on the serial interfaces connecting the routers. This ensured that OSPF neighbors could only form adjacencies if they shared the same authentication configuration. Temporary neighbor down messages were observed during configuration, which is expected behavior while authentication settings are being applied.

Verification commands confirmed that OSPF authentication was successfully enabled. The **show ip ospf interface** command showed that message-digest authentication is active on the serial interfaces, and the **show ip ospf neighbor** command confirmed that all routers reached the **FULL** neighbor state. Finally, connectivity was tested using **ping** between multiple PCs across different networks, and the results confirmed successful communication, proving that OSPF routing with authentication is working correctly.

```
R2#en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#key chain MIU_SEC
R2(config-keychain)#key 10
R2(config-keychain-key)#key-string MIU_AI_24
R2(config-keychain-key)#exit
R2(config-keychain)#int s0/1/1
R2(config-if)#ip ospf authentication message-digest
R2(config-if)#ip ospf message-digest-key 10 md5 MIU_AI_24
R2(config-if)#exit
R2(config)#
01:47:50: %OSPF-5-ADJCHG: Process 10, Nbr 209.165.201.1 on Serial0/1/1 from LOADING to FULL,
Loading Done
```

```
User Access Verification

Password:

R2>en
Password:
```

```
ISP>en
ISP#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
ISP(config)#key chain MIU_SEC
ISP(config-keychain)#key 10
ISP(config-keychain-key)#exit
ISP(config-keychain)#int s0/1/0
ISP(config-if)#ip ospf authentication message-digest
ISP(config-if)#ip ospf message-digest-key 10 md5 MIU_AI_24
ISP(config-if)#exit
ISP(config)#int s0/1/1
ISP(config-if)#
01:29:49: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.10.1 on Serial0/1/0 from LOADING to FULL,
Loading Done

ISP(config-if)#ip ospf authentication message-digest
ISP(config-if)#ip ospf message-digest-key 10 md5 MIU_AI_24
ISP(config-if)#exit
ISP(config)#
01:30:29: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.30.1 on Serial0/1/1 from FULL to DOWN,
Neighbor Down: Dead timer expired

01:30:29: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.30.1 on Serial0/1/1 from FULL to DOWN,
Neighbor Down: Interface down or detached
```

```
R1>show ip ospf interface

Serial0/1/0 is up, line protocol is up
  Internet address is 10.1.1.1/30, Area 0
  Process ID 10, Router ID 192.168.10.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 209.165.201.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 10
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 192.168.10.1/24, Area 0
  Process ID 10, Router ID 192.168.10.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.10.1, Interface address 192.168.10.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
R1>
```

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
R2>show ip ospf interface

Serial0/1/1 is up, line protocol is up
  Internet address is 10.2.2.1/30, Area 0
  Process ID 10, Router ID 192.168.30.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 209.165.201.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 10
GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.20.1/24, Area 0
  Process ID 10, Router ID 192.168.30.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.30.1, Interface address 192.168.20.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 192.168.30.1/24, Area 0
  Process ID 10, Router ID 192.168.30.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.30.1, Interface address 192.168.30.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Copy Paste

Top

ISP

Physical Config CLI Attributes

IOS Command Line Interface

```
%SYS-5-CONFIG_I: Configured from console by console

ISP#show ip ospf interface

Serial0/1/0 is up, line protocol is up
  Internet address is 10.1.1.2/30, Area 0
  Process ID 10, Router ID 209.165.201.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:01
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.10.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 10
Serial0/1/1 is up, line protocol is up
  Internet address is 10.2.2.2/30, Area 0
  Process ID 10, Router ID 209.165.201.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.30.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 10
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 209.165.201.1/29, Area 0
  Process ID 10, Router ID 209.165.201.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 209.165.201.1, Interface address 209.165.201.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
ISP#
```

Copy Paste

Top

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.20: bytes=32 time=2ms TTL=125
Reply from 192.168.20.20: bytes=32 time=2ms TTL=125
Reply from 192.168.20.20: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125
Reply from 192.168.20.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\>
```

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=26ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=15ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 26ms, Average = 11ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.11: bytes=32 time=2ms TTL=125
Reply from 192.168.10.11: bytes=32 time=2ms TTL=125
Reply from 192.168.10.11: bytes=32 time=13ms TTL=125

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 5ms

C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.12: bytes=32 time=2ms TTL=125
Reply from 192.168.10.12: bytes=32 time=21ms TTL=125
Reply from 192.168.10.12: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 21ms, Average = 8ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128
Reply from 192.168.20.10: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time=26ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=13ms TTL=125
Reply from 192.168.10.10: bytes=32 time=27ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 27ms, Average = 17ms

C:\>ping 192.168.20.20

Pinging 192.168.20.20 with 32 bytes of data:

Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128
Reply from 192.168.20.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.20.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

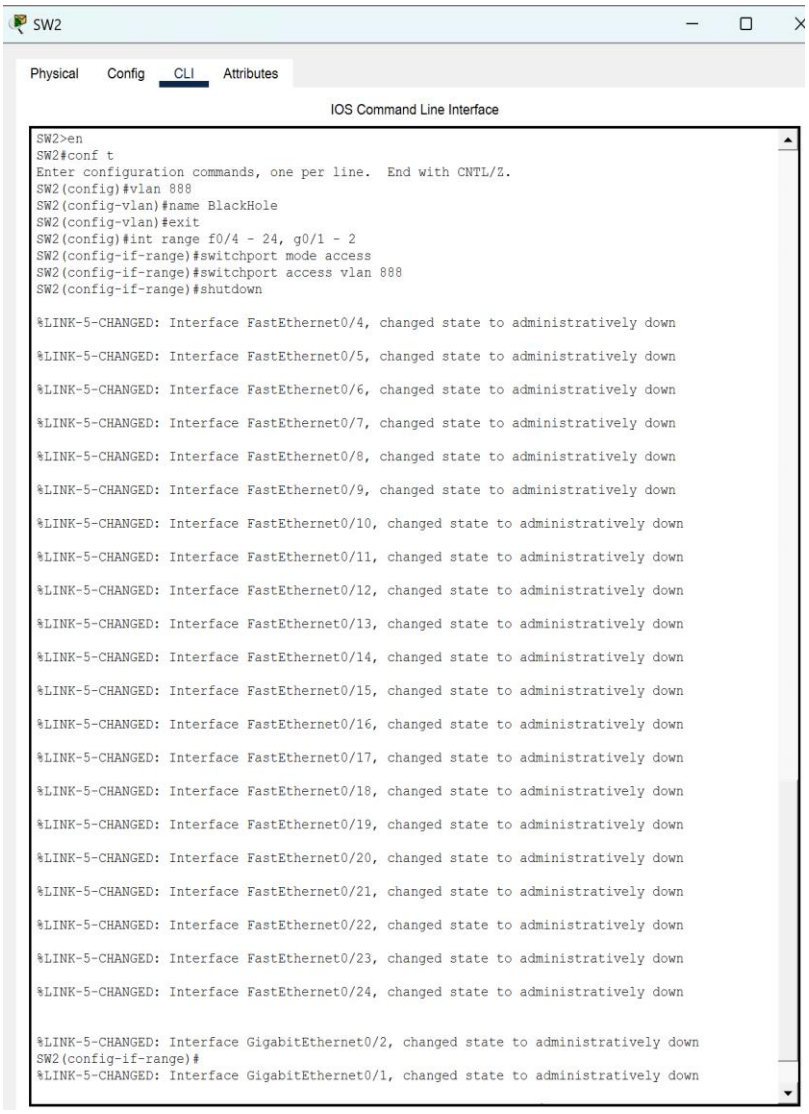
C:\>
```

Part 5: Switch Security Configuration

In this part, Layer 2 security features were implemented on switches SW1 and SW2 to improve network security and prevent unauthorized access.

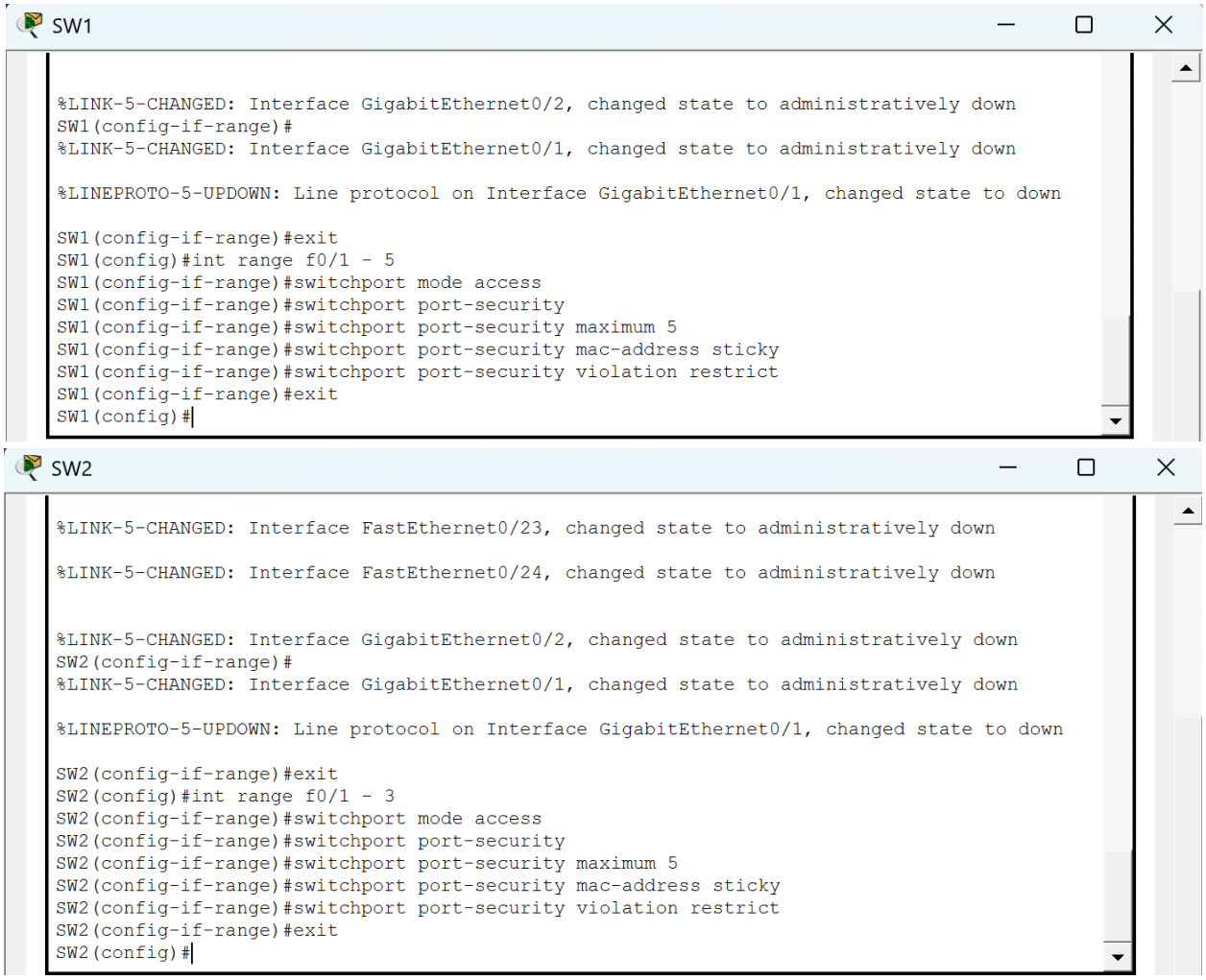
First, all unused switch ports on both switches were secured. A dedicated VLAN with ID **888** named **BlackHole** was created on SW1 and SW2. All unused FastEthernet and GigabitEthernet ports were configured as access ports, assigned to the BlackHole VLAN, and administratively shut down. This ensures that unused ports cannot be exploited by unauthorized devices.



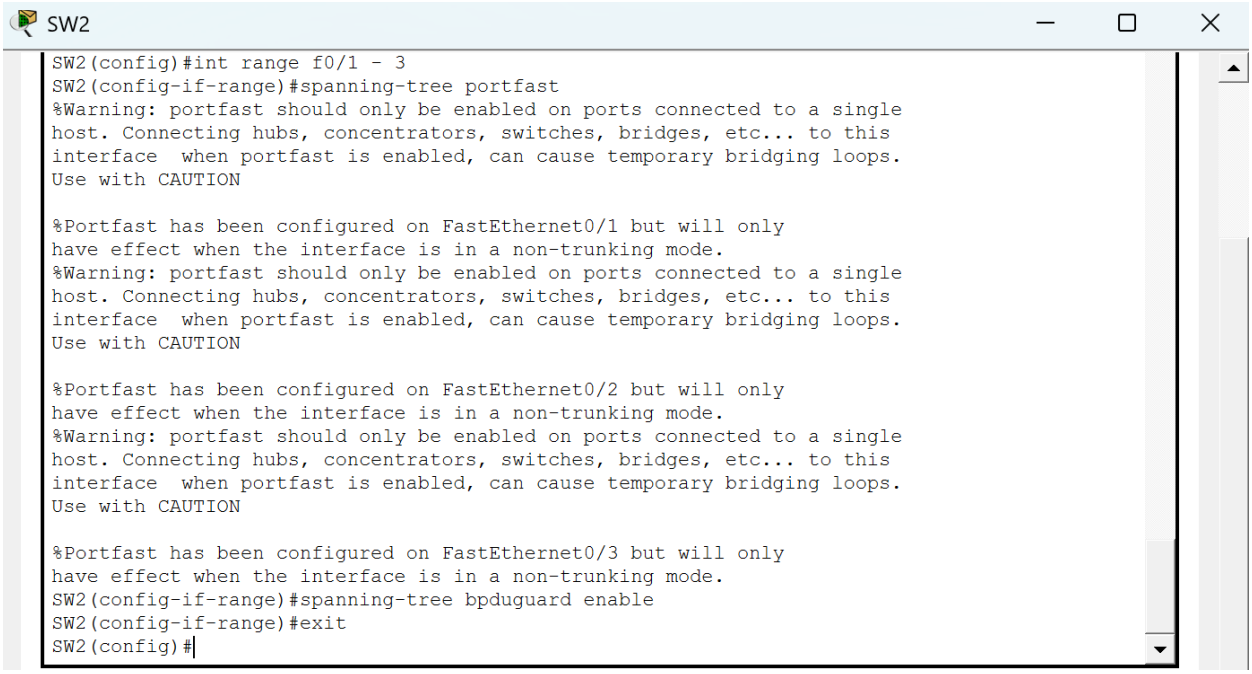


Next, **port security** was configured on all active access ports on both switches. Port security was enabled with a maximum of **5 MAC addresses** allowed per port. Sticky MAC

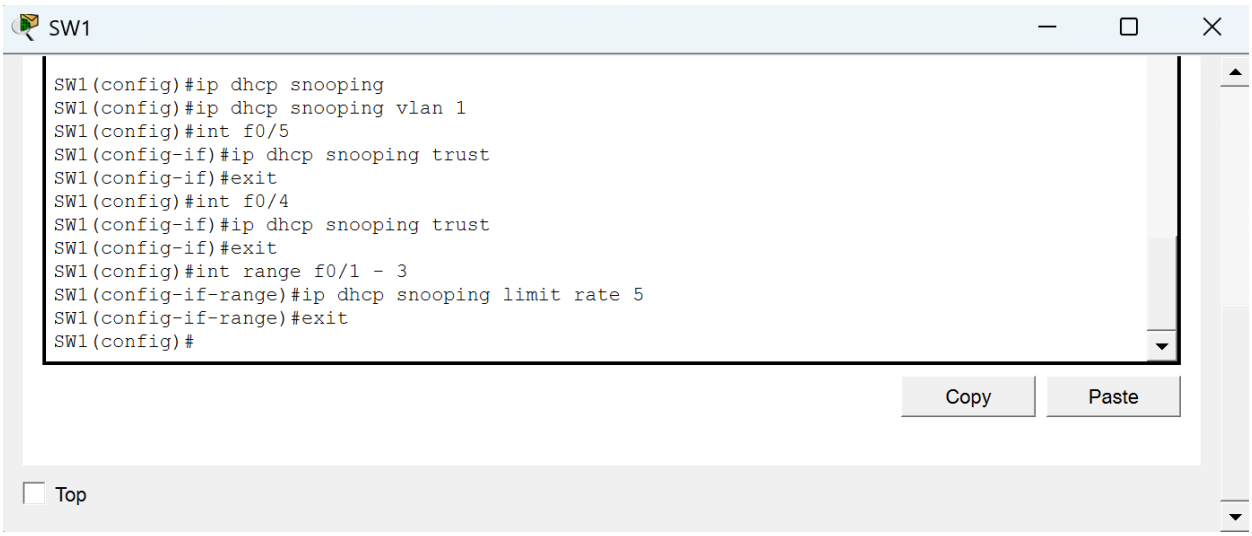
address learning was used so that learned MAC addresses are automatically added to the running configuration. The violation mode was set to **restrict**, which drops packets from unknown MAC addresses and generates Syslog messages without shutting down the port.



After that, **Spanning Tree security** features were applied on SW2. PortFast was enabled on all active access ports to allow end devices to connect quickly without STP delays. In addition, **BPDU Guard** was enabled to protect the network from accidental or malicious switch connections by disabling ports that receive BPDU packets.



Finally, **DHCP Snooping** was configured on SW1 to protect against rogue DHCP servers. DHCP snooping was enabled for VLAN 1. The ports connected to the router and the legitimate DHCP server were configured as **trusted ports**, while all untrusted access ports were rate-limited to **5 DHCP packets per second**. This configuration helps prevent DHCP starvation and spoofing attacks.



Overall, these configurations successfully enhanced Layer 2 security by controlling port access, protecting against MAC flooding, STP manipulation, and DHCP-based attacks.

Part 6: Configure a Site-to-Site IPsec VPN

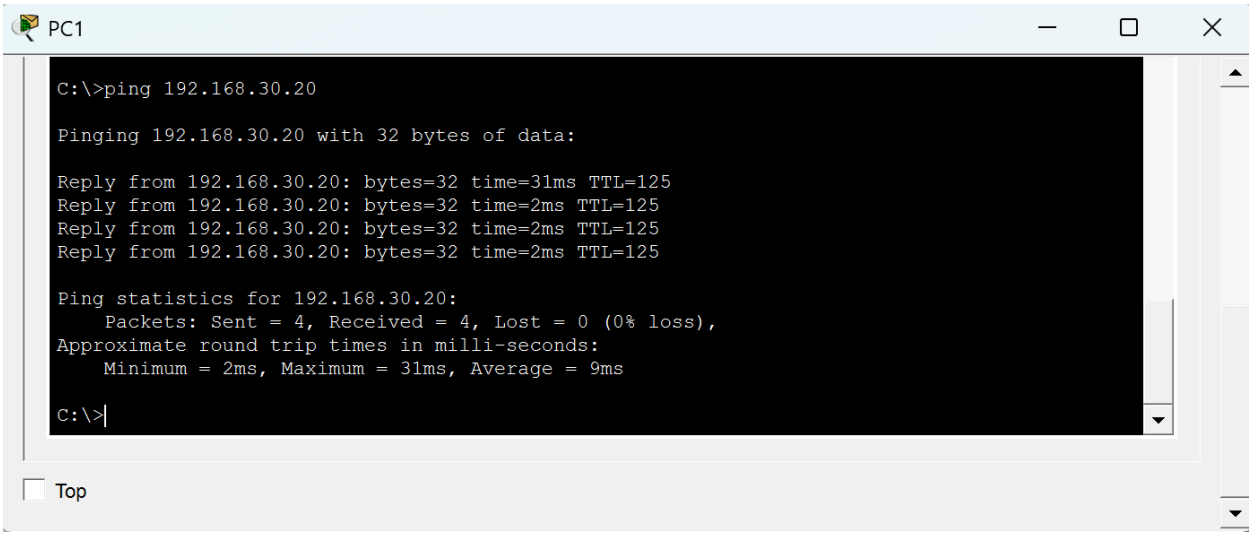
Overview

In this part, a site-to-site IPsec VPN was configured between **R1 and R2** to secure traffic between their LANs while passing through the ISP, which acts only as a transit network. The VPN ensures that sensitive traffic between the two sites is encrypted and authenticated over the untrusted network.

Step 1: Test Connectivity Before VPN

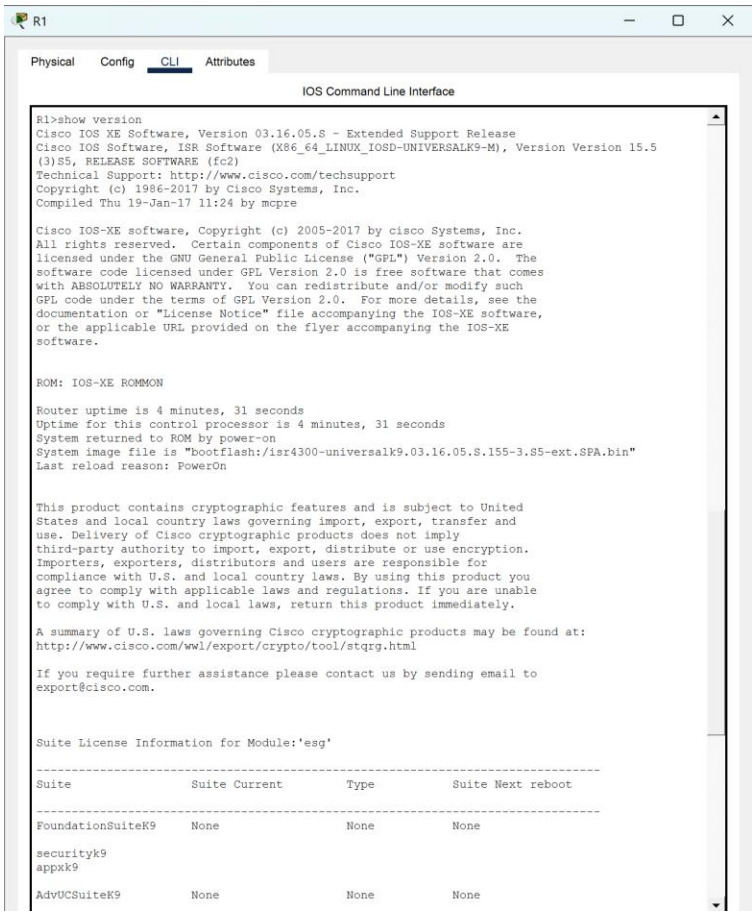
Before configuring the VPN, connectivity between the LANs was tested to ensure that routing is working correctly.

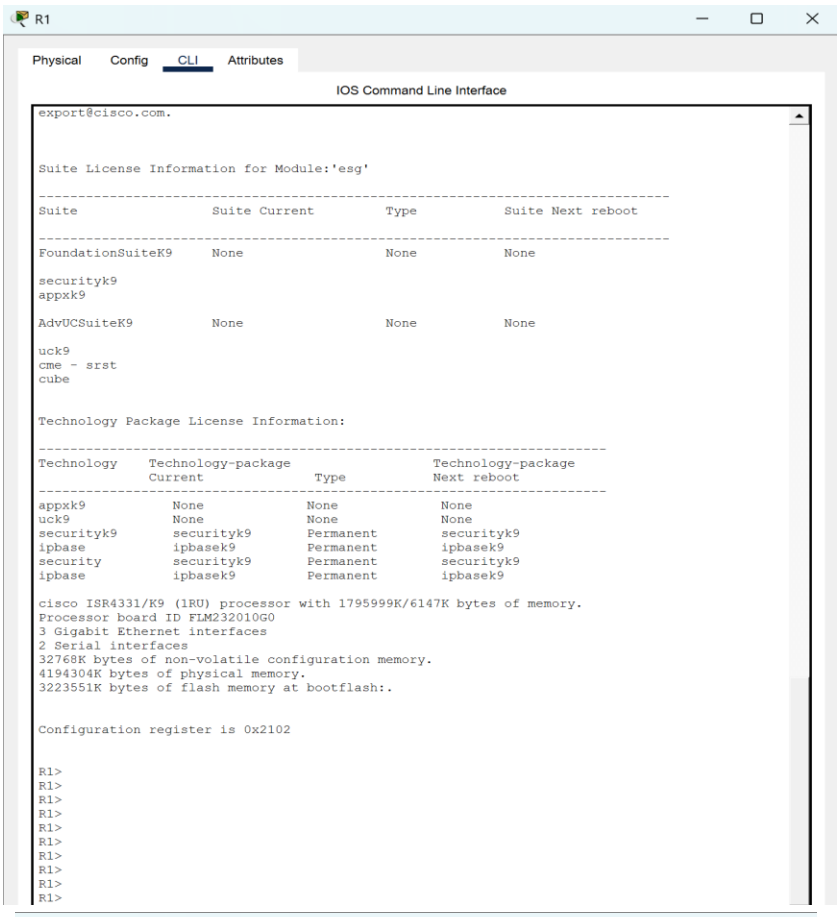
Initial connectivity was verified by pinging the Guest PC from PC1. Since OSPF routing was already configured, the ping was successful, confirming end-to-end reachability before enabling the VPN.

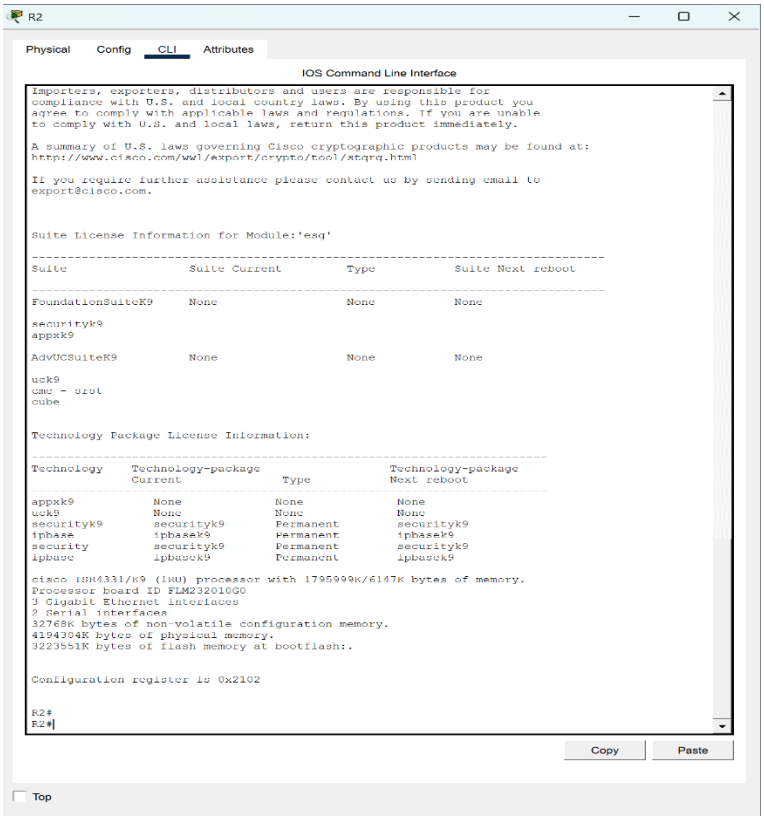


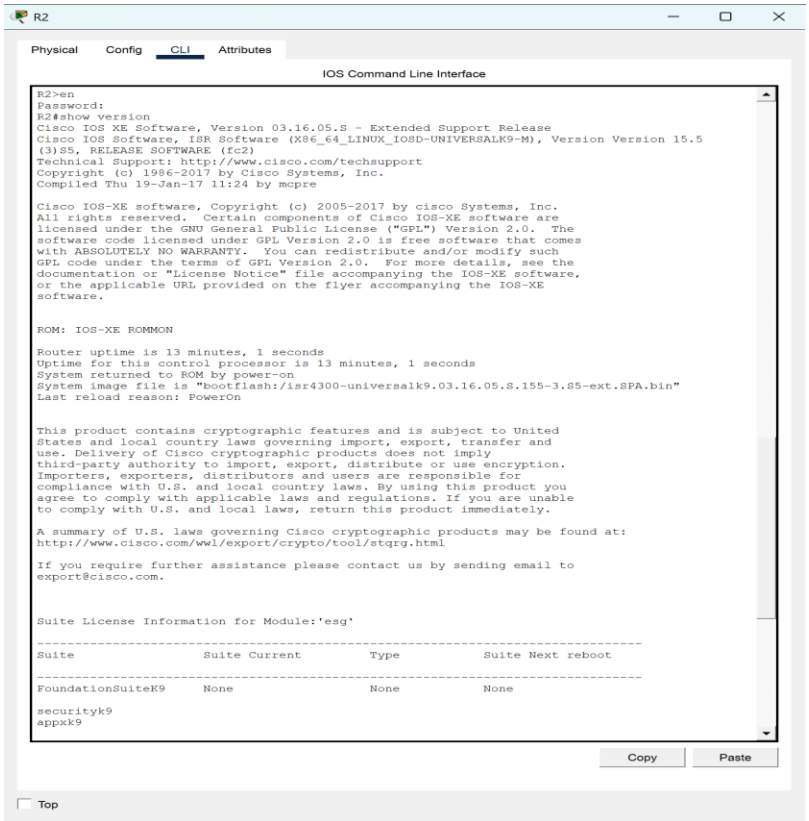
Step 2: Enable Security Technology Package

The Security Technology package was enabled on both R1 and R2. The router was reloaded after accepting the license agreement, and the configuration was verified using the show version command.


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
R1>show version
Cisco IOS XE Software, Version 03.16.05.S - Extended Support Release
Cisco IOS Software, ISR Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version Version 15.5
(3)S5, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2017 by Cisco Systems, Inc.
Compiled Thu 19-Jan-17 11:24 by mcpre
Cisco IOS-XE software, Copyright (c) 2005-2017 by cisco Systems, Inc.
All rights reserved. Certain components of Cisco IOS-XE software are
licensed under the GNU General Public License ("GPL") Version 2.0. The
software code licensed under GPL Version 2.0 is free software that comes
with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
or the applicable URL provided on the flyer accompanying the IOS-XE
software.
ROM: IOS-XE ROMMON
Router uptime is 4 minutes, 31 seconds
Uptime for this control processor is 4 minutes, 31 seconds
System returned to ROM by power-on
System image file is "bootflash:/isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin"
Last reload reason: PowerOn
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Suite License Information for Module:'esg'
Suite
Suite Current
Type
Suite Next reboot
FoundationSuiteK9
None
None
None
securityk9
appxk9
AdvUCSuiteK9
None
None
None


R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
export@cisco.com.
Suite License Information for Module:'esg'
Suite
Suite Current
Type
Suite Next reboot
FoundationSuiteK9
None
None
None
securityk9
appxk9
AdvUCSuiteK9
None
None
None
uck9
cme - srst
cube
Technology Package License Information:
Technology
Technology-package Current
Type
Technology-package Next reboot
appxk9
None
None
None
uck9
None
None
None
securityk9
securityk9
Permanent
securityk9
ipbase
ipbasek9
Permanent
ipbasek9
security
securityk9
Permanent
securityk9
ipbase
ipbasek9
Permanent
ipbasek9
cisco ISR4331/K9 (1RU) processor with 1795999K/6147K bytes of memory.
Processor board ID FLM232010G0
3 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.
Configuration register is 0x2102
R1>
R1>
R1>
R1>
R1>
R1>
R1>
R1>
R1>
R1>
R1>


R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Suite License Information for Module:'esg'
Suite
Suite Current
Type
Suite Next reboot
FoundationSuiteK9
None
None
None
securityk9
appxk9
AdvUCSuiteK9
None
None
None
uck9
cme - srst
cube
Technology Package License Information:
Technology
Technology-package Current
Type
Technology-package Next reboot
appxk9
None
None
None
uck9
None
None
None
securityk9
securityk9
Permanent
securityk9
ipbase
ipbasek9
Permanent
ipbasek9
security
securityk9
Permanent
securityk9
ipbase
ipbasek9
Permanent
ipbasek9
cisco ISR4331/K9 (1RU) processor with 1795999K/6147K bytes of memory.
Processor board ID FLM232010G0
3 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.
Configuration register is 0x2102
R2#
R2#
Copy
Paste
Top


R2
Physical
Config
CLI
Attributes
IOS Command Line Interface
R2>en
Password:
R2#show version
Cisco IOS XE Software, Version 03.16.05.S - Extended Support Release
Cisco IOS Software, ISR Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version Version 15.5
(3)S5, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2017 by Cisco Systems, Inc.
Compiled Thu 19-Jan-17 11:24 by mcpre
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with ABSOLUTELY NO WARRANTY. You can redistribute and/or modify such
GPL code under the terms of GPL Version 2.0. For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
or the applicable URL provided on the flyer accompanying the IOS-XE
software.
ROM: IOS-XE ROMMON
Router uptime is 13 minutes, 1 seconds
Uptime for this control processor is 13 minutes, 1 seconds
System returned to ROM by power-on
System image file is "bootflash:/isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin"
Last reload reason: PowerOn
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Suite License Information for Module:'esg'
Suite
Suite Current
Type
Suite Next reboot
FoundationSuiteK9
None
None
None
securityk9
appxk9
Copy
Paste
Top

Step 3: Identify Interesting Traffic Using ACL

An access control list was configured to define the traffic that should be encrypted by the VPN.

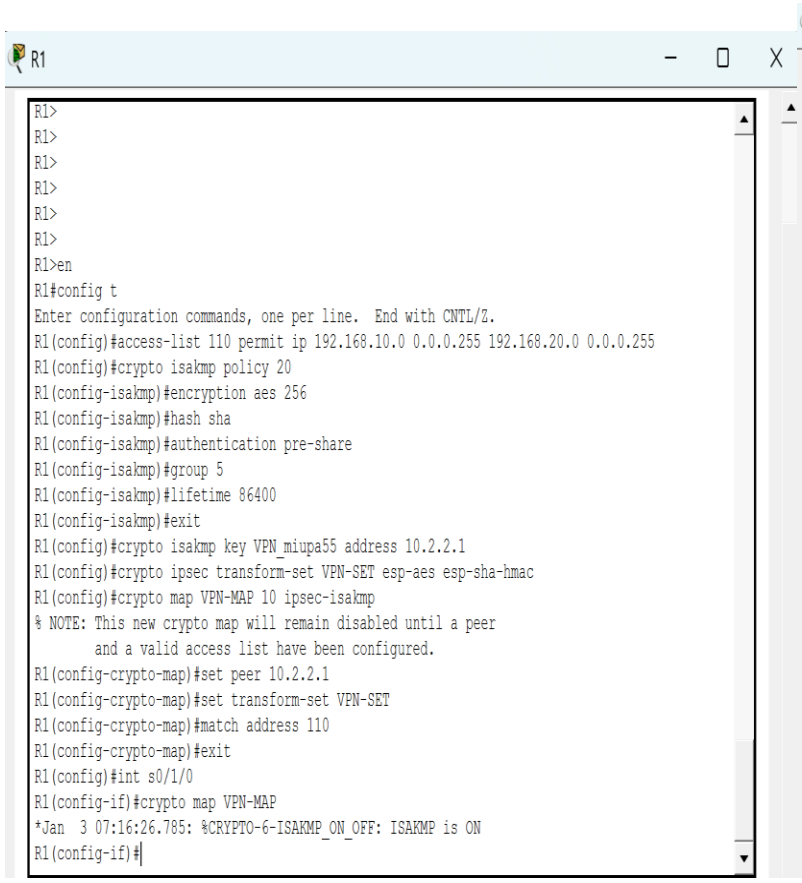
ACL 110 was configured on R1 and R2 to identify interesting traffic between the LAN of R1 (192.168.10.0/24) and the LAN of R2 (192.168.20.0/24). Only this traffic will be encrypted by the VPN tunnel.

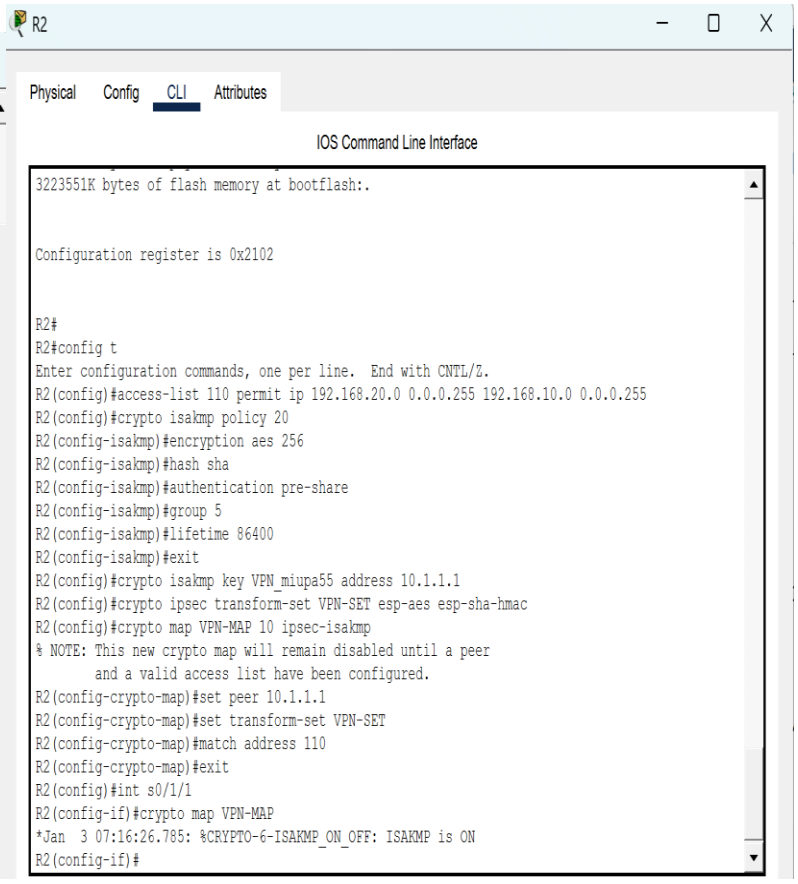
Step 4: Configure IKE Phase 1 (ISAKMP Policy)

ISAKMP Phase 1 was configured on both routers using AES-256 encryption, SHA-1 hashing, pre-shared key authentication, and Diffie-Hellman group 5. The shared key VPN_miupa55 was configured on both peers.

Step 5: Configure IKE Phase 2 (IPsec Policy)

An IPsec transform set named **VPN-SET** was created using ESP-AES for encryption and ESP-SHA-HMAC for authentication. A crypto map named **VPN-MAP** was then configured and bound to the outgoing serial interfaces on both routers.





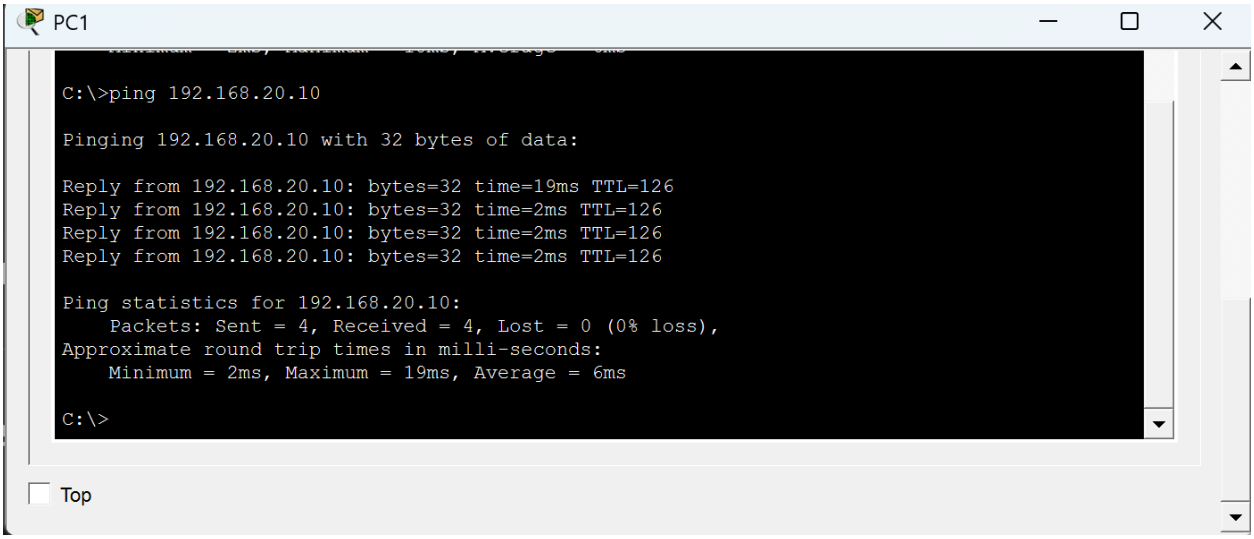
Step 6: Verify IPsec Tunnel Before Traffic

Before generating interesting traffic, the show crypto ipsec sa command showed that no packets were encrypted or decrypted, indicating that the tunnel had not yet been triggered.



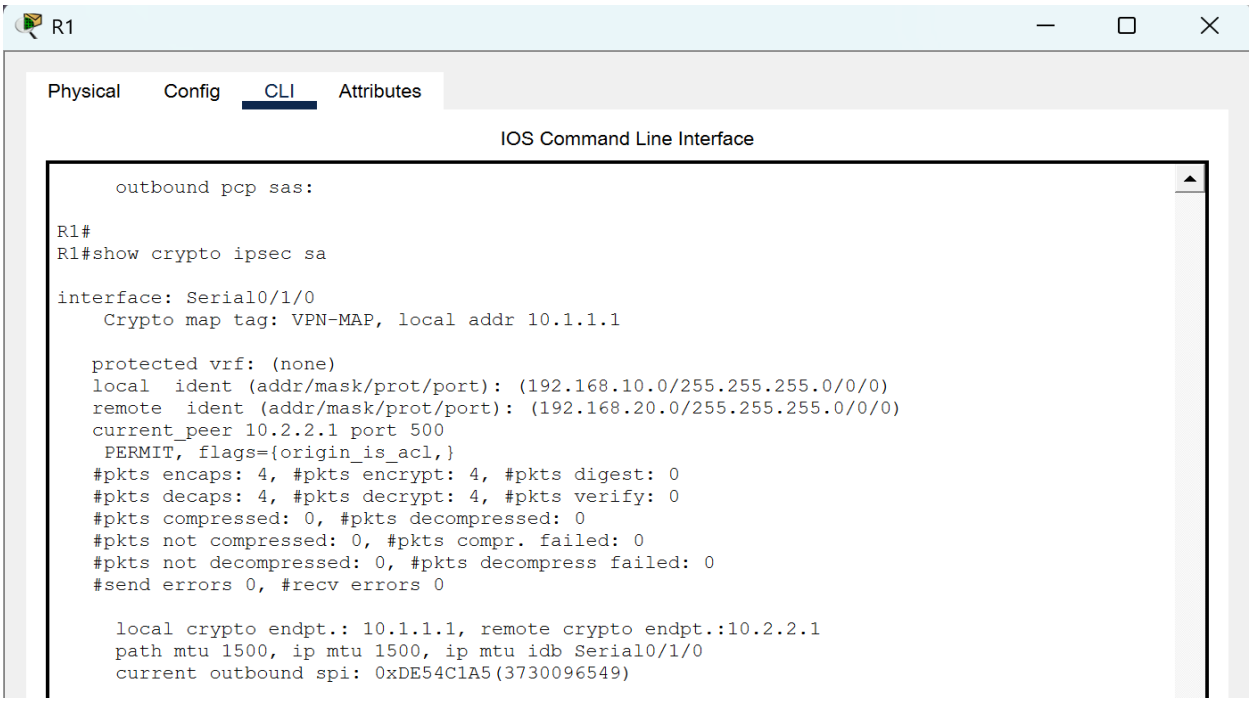
Step 7: Generate Interesting Traffic

Interesting traffic was generated by pinging PC5 from PC1. This traffic matched ACL 110 and triggered the IPsec VPN tunnel.



Step 8: Verify IPsec Tunnel After Traffic

After generating traffic, the show crypto ipsec sa command showed an increase in encrypted and decrypted packets, confirming that the IPsec VPN tunnel was successfully established and operational.



The site-to-site IPsec VPN between R1 and R2 was successfully configured and verified. Secure communication was established between the two LANs, with only defined traffic being encrypted. Testing confirmed correct VPN operation, while uninteresting traffic remained unencrypted.

Part 7: Configuring a Zone-Based Policy Firewall (ZPF) on R2

Objective

The objective of this part is to configure a **Zone-Based Policy Firewall (ZPF)** on router **R2** to control traffic between the **INSIDE**, **GUEST**, and **OUTSIDE** networks based on security requirements.
The firewall ensures secure access to external resources while preventing unauthorized traffic from entering trusted internal networks.

Security Design Overview

- The **INSIDE network** is considered **trusted** and is allowed to initiate all types of traffic (TCP, UDP, ICMP) toward the OUTSIDE.
- The **GUEST network** is considered **untrusted** and is restricted to **web-based traffic only** (HTTP, HTTPS, and DNS) toward the OUTSIDE.
- **No traffic initiated from the OUTSIDE** is allowed to access INSIDE or GUEST networks **except VPN traffic**, which is handled separately.

This design follows the **principle of least privilege** and reflects a real enterprise security model.

Step 1: ZPF Configuration for INSIDE to OUTSIDE

In this step, two security zones were created:

- **INSIDE**
- **OUTSIDE**

A class map named **INSIDE-PROTOCOLS** was defined to match **TCP, UDP, and ICMP traffic**, allowing full communication from the trusted INSIDE network.

A policy map named **INSIDE-TO-OUTSIDE-PM** was then created and bound to this class map, with matched traffic being **inspected**.

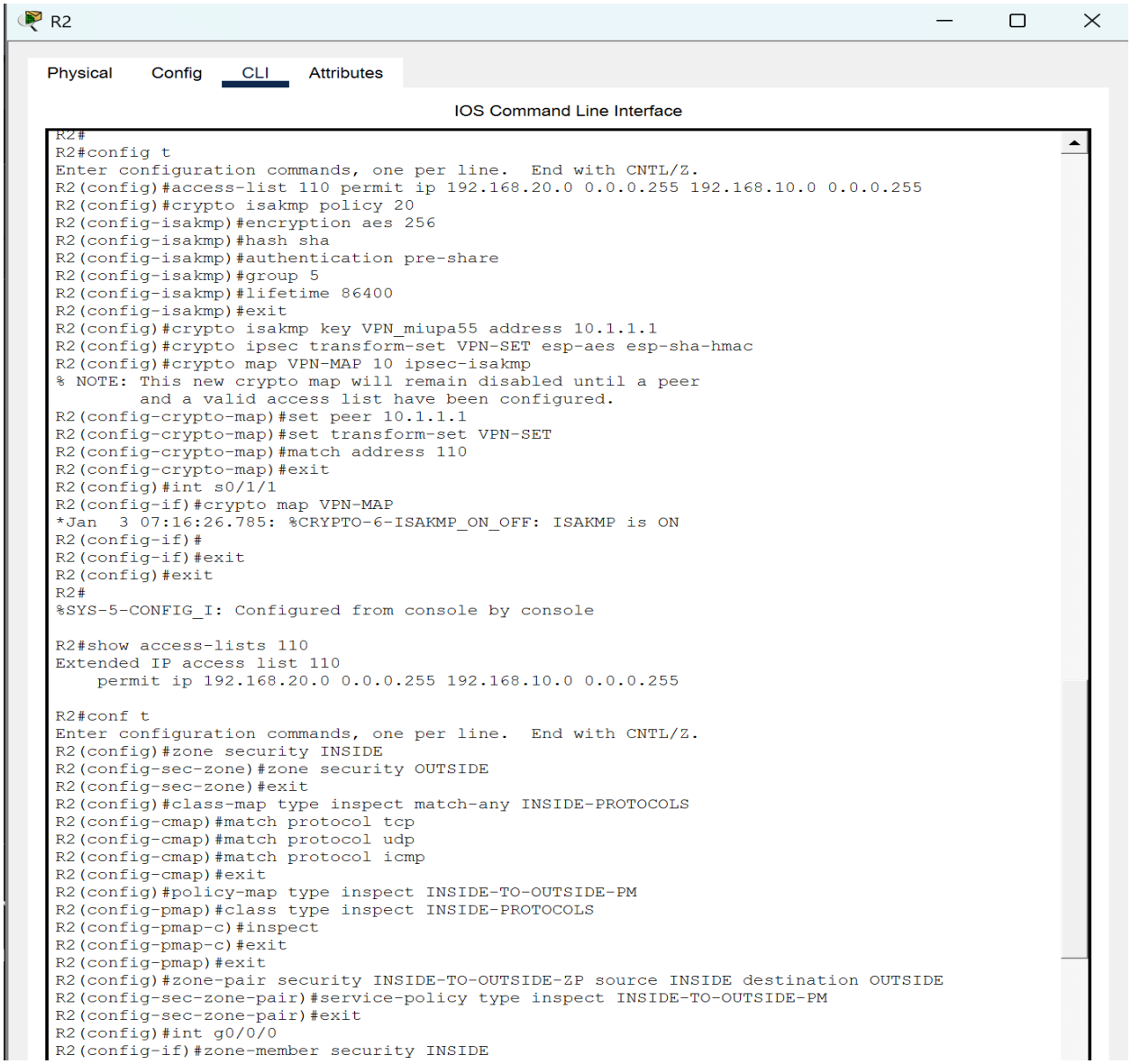
Inspection allows return traffic automatically without explicitly permitting it in the opposite direction.

A zone pair named **INSIDE-TO-OUTSIDE-ZP** was created to enforce this policy between the INSIDE and OUTSIDE zones.

Finally, interfaces were assigned to their respective zones:

- **G0/0/0 → INSIDE**
- **S0/1/1 → OUTSIDE**

This ensures that all traffic leaving the INSIDE network toward the OUTSIDE is inspected and allowed.



Step 2: ZPF Configuration for GUEST to OUTSIDE

A separate security zone named **GUEST** was created to isolate untrusted users.

A class map called **GUEST-PROTOCOLS** was configured to allow only:

- **HTTP**
- **HTTPS**
- **DNS**

This restricts guest users to basic internet access only.

A policy map named **GUEST-TO-OUTSIDE-PM** was created and configured to inspect matched traffic.
A zone pair named **GUEST-TO-OUTSIDE-ZP** was then applied to enforce the policy between the GUEST and OUTSIDE zones.

The **G0/0/1** interface was assigned to the **GUEST** zone, ensuring that all guest traffic is filtered according to the defined rules.



Step 3: ZPF Configuration for OUTSIDE to INSIDE (VPN Only)

To prevent unauthorized access from the OUTSIDE network, a strict policy was implemented.

A named ACL **REMOTE-TRAFFIC** was created to permit **only VPN traffic** between:

- **192.168.10.0/24** (R1 INSIDE)
- **192.168.20.0/24** (R2 INSIDE)

This ACL was used inside a class map named **OUTSIDE-TRAFFIC** with a **match-all** condition.

A policy map named **OUTSIDE-TO-INSIDE-PM** was created to inspect only this VPN traffic.

A zone pair named **OUTSIDE-TO-INSIDE-ZP** was then applied, ensuring that **no other traffic from OUTSIDE can access INSIDE**, except through the VPN tunnel.

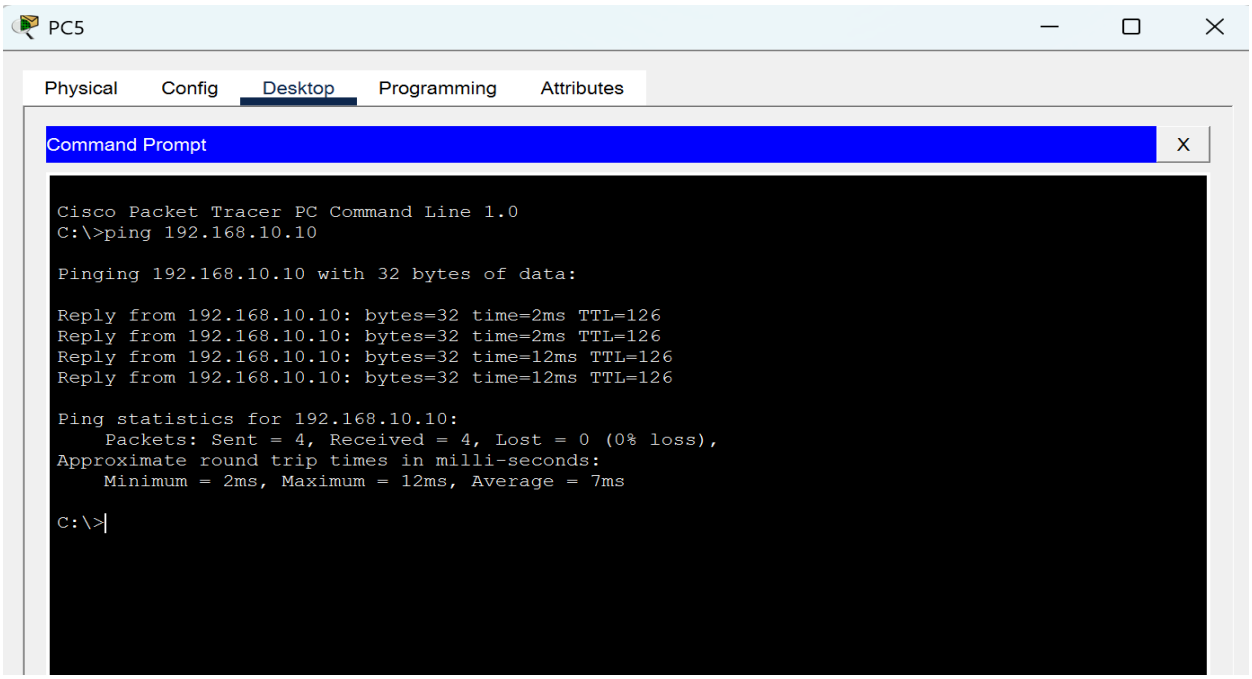
```
R2(config)#class-map type inspect match-all OUTSIDE-TRAFFIC
R2(config-cmap)#match access-group name REMOTE-TRAFFIC
R2(config-cmap)#exit
R2(config)#policy-map type inspect OUTSIDE-TO-INSIDE-PM
R2(config-pmap)#class type inspect OUTSIDE-TRAFFIC
R2(config-pmap-c)#inspect
%No specific protocol configured in class OUTSIDE-TRAFFIC for inspection. All protocols will
be inspected
R2(config-pmap-c)#exit
R2(config-pmap)#exit
R2(config)#zone-pair security OUTSIDE-TO-INSIDE-ZP source OUTSIDE destination INSIDE
R2(config-sec-zone-pair)#service-policy type inspect OUTSIDE-TO-INSIDE-PM
R2(config-sec-zone-pair)#
```

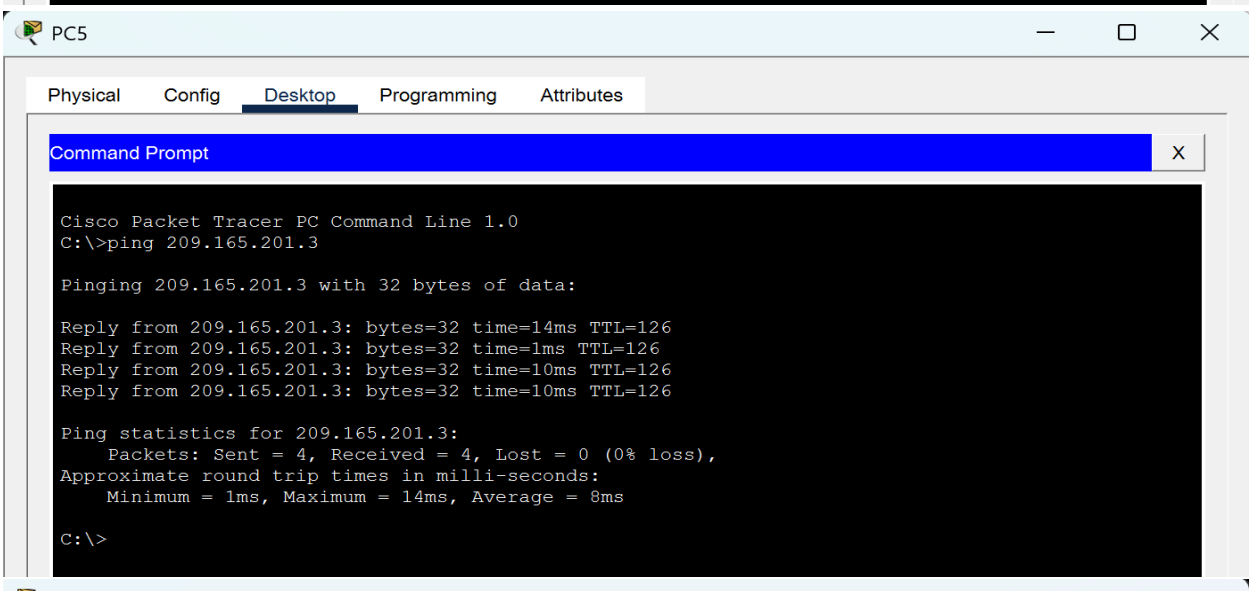
Step 4: Verification

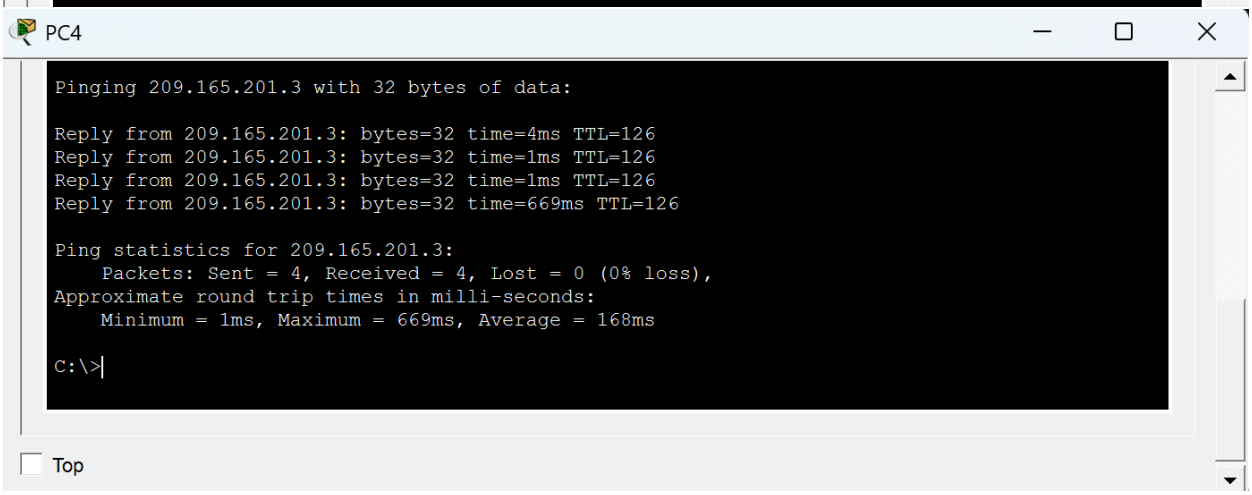
The following verifications were successfully performed:

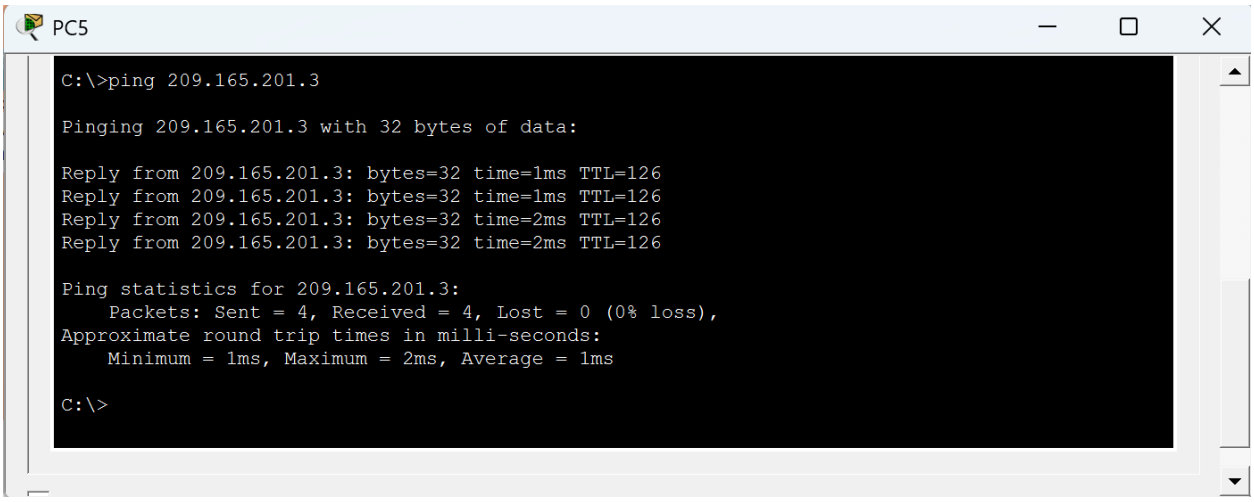
- PCs in the **INSIDE** network were able to access **miu.edu.eg**
- The **VPN tunnel between PC1 and PC5** was confirmed operational
- **Guest PCs** could browse the web but were restricted from accessing internal networks
- All non-VPN traffic initiated from the OUTSIDE toward INSIDE was blocked

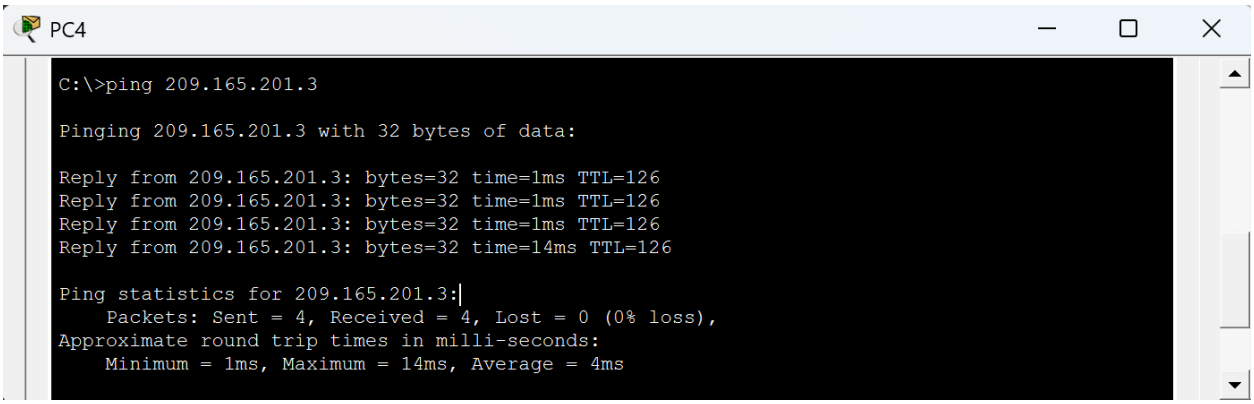
These results confirm that the ZPF configuration is functioning correctly and meets the security requirements.

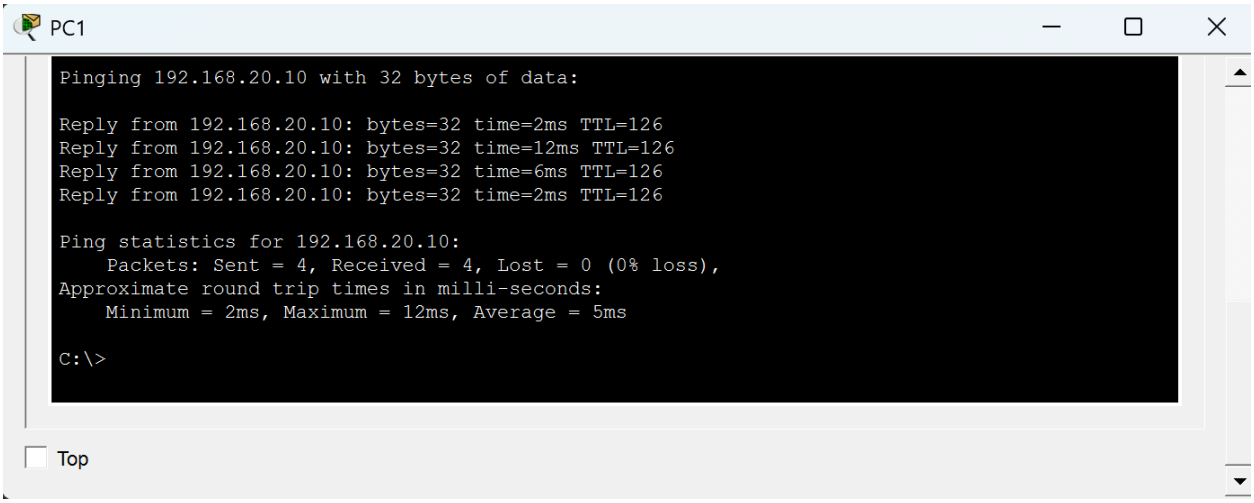

PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10
Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time=2ms TTL=126
Reply from 192.168.10.10: bytes=32 time=2ms TTL=126
Reply from 192.168.10.10: bytes=32 time=12ms TTL=126
Reply from 192.168.10.10: bytes=32 time=12ms TTL=126
Ping statistics for 192.168.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 12ms, Average = 7ms
C:\>


PC5
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 209.165.201.3
Pinging 209.165.201.3 with 32 bytes of data:
Reply from 209.165.201.3: bytes=32 time=14ms TTL=126
Reply from 209.165.201.3: bytes=32 time=1ms TTL=126
Reply from 209.165.201.3: bytes=32 time=10ms TTL=126
Reply from 209.165.201.3: bytes=32 time=10ms TTL=126
Ping statistics for 209.165.201.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 14ms, Average = 8ms
C:\>


PC4
Pinging 209.165.201.3 with 32 bytes of data:
Reply from 209.165.201.3: bytes=32 time=4ms TTL=126
Reply from 209.165.201.3: bytes=32 time=1ms TTL=126
Reply from 209.165.201.3: bytes=32 time=1ms TTL=126
Reply from 209.165.201.3: bytes=32 time=669ms TTL=126
Ping statistics for 209.165.201.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 669ms, Average = 168ms
C:\>
Top







The Zone-Based Policy Firewall on R2 was successfully implemented to enforce network segmentation and traffic control.

Trusted users are granted full access, guest users are restricted, and external threats are mitigated while still allowing secure VPN connectivity.

This configuration enhances overall network security while maintaining necessary functionality.

```
R2#show run | section zone
zone security INSIDE
zone security OUTSIDE
zone security GUEST
zone-pair security INSIDE-TO-OUTSIDE-ZP source INSIDE destination OUTSIDE
 service-policy type inspect INSIDE-TO-OUTSIDE-PM
zone-pair security GUEST-TO-OUTSIDE-ZP source GUEST destination OUTSIDE
 service-policy type inspect GUEST-TO-OUTSIDE-PM
zone-pair security OUTSIDE-TO-INSIDE-ZP source OUTSIDE destination INSIDE
 service-policy type inspect OUTSIDE-TO-INSIDE-PM
 zone-member security INSIDE
 zone-member security GUEST
 zone-member security OUTSIDE
R2#show run | include zone security | zone-member | zone-pair
zone security INSIDE
zone security OUTSIDE
zone security GUEST
 zone-member security INSIDE
 zone-member security GUEST
 zone-member security OUTSIDE
```

```
R2#show run | section class-map
class-map type inspect match-any INSIDE-PROTOCOLS
 match protocol tcp
 match protocol udp
 match protocol icmp
class-map type inspect match-any GUEST-PROTOCOLS
 match protocol http
 match protocol https
 match protocol dns
class-map type inspect match-all OUTSIDE-TRAFFIC
 match access-group name REMOTE-TRAFFIC
R2#show run | section policy-map type inspect
policy-map type inspect INSIDE-TO-OUTSIDE-PM
 class type inspect INSIDE-PROTOCOLS
  inspect
policy-map type inspect GUEST-TO-OUTSIDE-PM
 class type inspect GUEST-PROTOCOLS
  inspect
policy-map type inspect OUTSIDE-TO-INSIDE-PM
 class type inspect OUTSIDE-TRAFFIC
  inspect
R2#show run | section ^zone-pair security|service policy
zone-pair security INSIDE-TO-OUTSIDE-ZP source INSIDE destination OUTSIDE
 service-policy type inspect INSIDE-TO-OUTSIDE-PM
zone-pair security GUEST-TO-OUTSIDE-ZP source GUEST destination OUTSIDE
 service-policy type inspect GUEST-TO-OUTSIDE-PM
zone-pair security OUTSIDE-TO-INSIDE-ZP source OUTSIDE destination INSIDE
 service-policy type inspect OUTSIDE-TO-INSIDE-PM
```

Part 8: Configuring a Stateless Firewall on R1 Using a Named Extended ACL

Objective

The objective of this part is to implement a **stateless firewall** on router **R1** using a **named extended access control list (ACL)** to selectively restrict traffic based on **source, destination, and protocol**.
The ACL enforces security policies for HTTP, HTTPS, FTP, and ICMP traffic originating from specific PCs.

Security Policy Requirements

The following policies were required:

1. Block **HTTP and HTTPS** access from **PC1** to **miu.edu.eg**.
2. Block **FTP** access from **PC2** to the **FTP Server**.
3. Block **ICMP (ping)** traffic from **PC3** to:
    - **miu.edu.eg**
    - **FTP Server**
4. Permit **all other traffic** not explicitly denied.

ACL Design Explanation

A **named extended ACL (BLOCK_TRAFFIC)** was used because extended ACLs allow filtering based on:

- Source IP address
- Destination IP address
- Protocol type (TCP, UDP, ICMP)
- Application ports (HTTP, HTTPS, FTP)

The ACL entries were written **in the exact order specified** in the instructions to ensure correct packet matching, since ACLs are processed **top-down**.

A final **permit ip any any** statement was added to explicitly allow all remaining traffic.

ACL Placement Justification

The traffic being filtered originates from the **192.168.10.0/24** network.
According to best practice, **extended ACLs should be applied closest to the source** of the traffic.

Therefore, the ACL was applied:

- On **R1**
- On the **interface connected to the 192.168.10.0/24 network**
- In the **inbound direction**

This minimizes unnecessary traffic traversal and improves efficiency.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list extended BLOCK_TRAFFIC
R1(config-ext-nacl)#deny tcp host 192.168.10.10 host 209.165.201.3 eq 80
R1(config-ext-nacl)#deny tcp host 192.168.10.10 eq 80 host 209.165.201.3
R1(config-ext-nacl)#deny tcp host 192.168.10.10 host 209.165.201.3 eq 443
R1(config-ext-nacl)#deny tcp host 192.168.10.10 eq 443 host 209.165.201.3
R1(config-ext-nacl)#deny tcp host 192.168.10.11 host 209.165.201.4 eq 21
R1(config-ext-nacl)#deny icmp host 192.168.10.12 host 209.165.201.3
R1(config-ext-nacl)#deny icmp host 192.168.10.12 host 209.165.201.4
R1(config-ext-nacl)#permit ip any any
R1(config-ext-nacl)#exit
R1(config)#show access-lists
```

R1

```
%SYS-5-CONFIG_I: Configured from console by console

R1#show access-lists
Extended IP access list 110
    10 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 (1 match(es))
Extended IP access list BLOCK_TRAFFIC
    10 deny tcp host 192.168.10.10 host 209.165.201.3 eq www
    20 deny tcp host 192.168.10.10 eq www host 209.165.201.3
    30 deny tcp host 192.168.10.10 host 209.165.201.3 eq 443
    40 deny tcp host 192.168.10.10 eq 443 host 209.165.201.3
    50 deny tcp host 192.168.10.11 host 209.165.201.4 eq ftp
    60 deny icmp host 192.168.10.12 host 209.165.201.3
    70 deny icmp host 192.168.10.12 host 209.165.201.4
    80 permit ip any any

R1#
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int g0/0/1
R1(config-if)#ip access-group BLOCK_TRAFFIC in
R1(config-if)#exit
R1(config)#
```

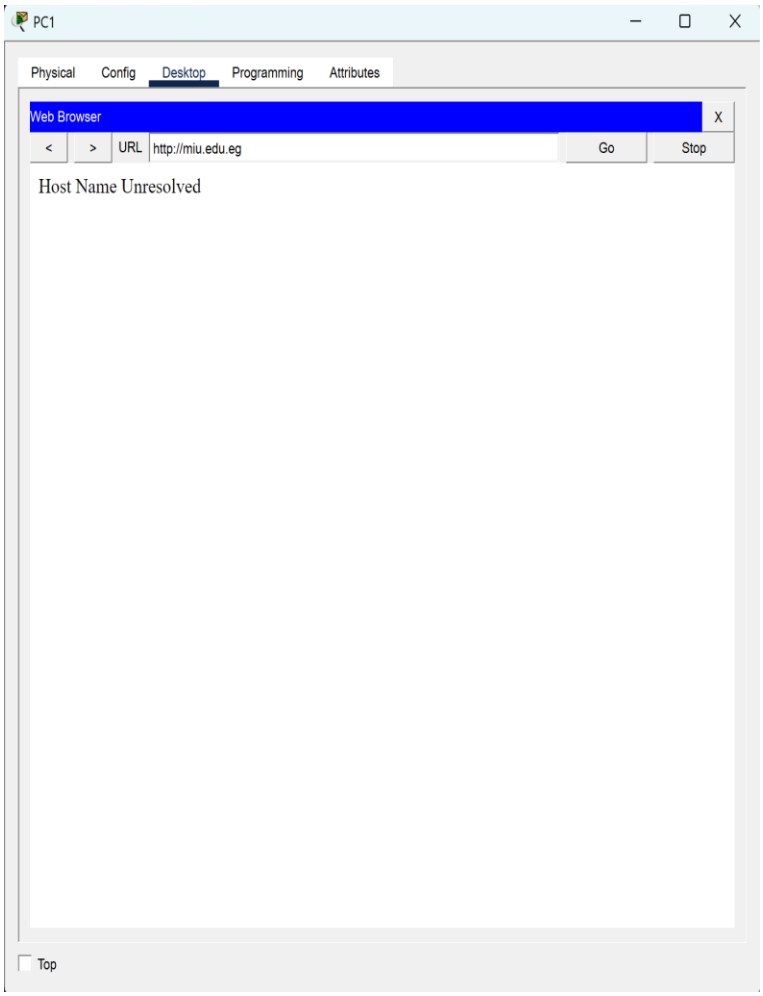
Verification and Testing

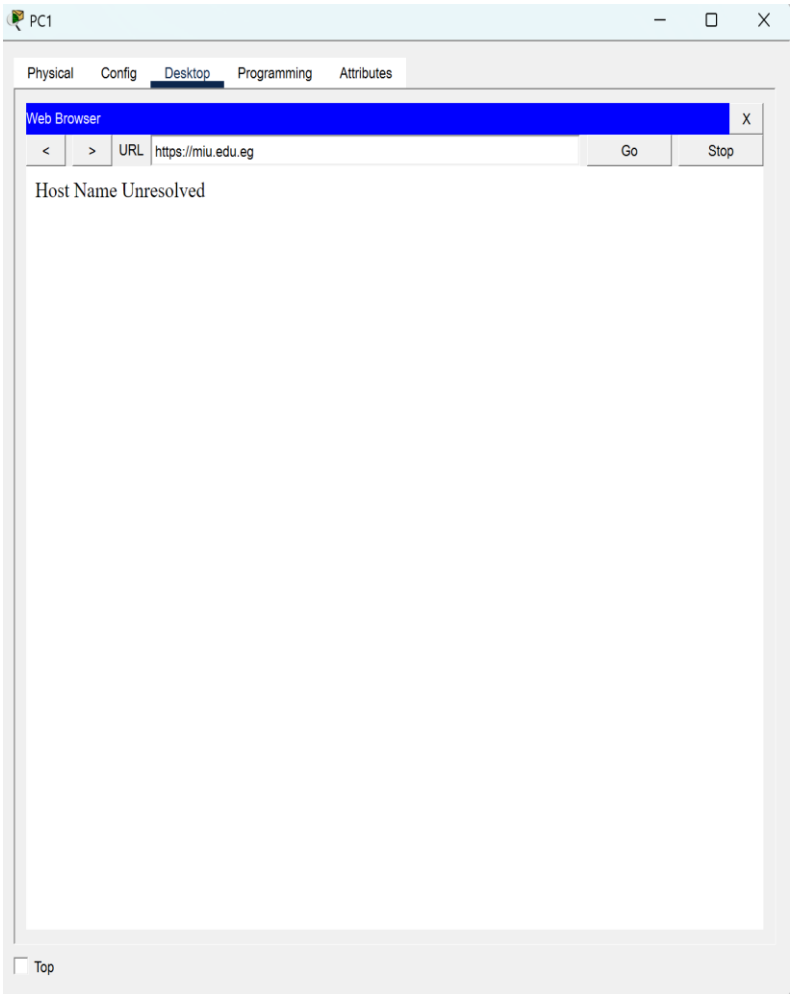
Before performing the following verification and testing steps, it should be noted that **Part 11 (Server Configuration)** is a required prerequisite.
The servers used in these tests (Miu.edu.eg Server and FTP Server) must be properly configured and operational to ensure accurate verification results.
Part 11 is explained and implemented later in this report, and the tests below assume that the required server services are correctly set up.

1. PC1 – HTTP/HTTPS Access Test

- PC1 attempted to access **miu.edu.eg** using both HTTP and HTTPS.
- Access was **successfully blocked**, confirming that the ACL correctly denied TCP traffic on ports **80 and 443**.
- ICMP traffic from PC1 remained permitted, as required.

This confirms correct enforcement of the HTTP/HTTPS deny rules.

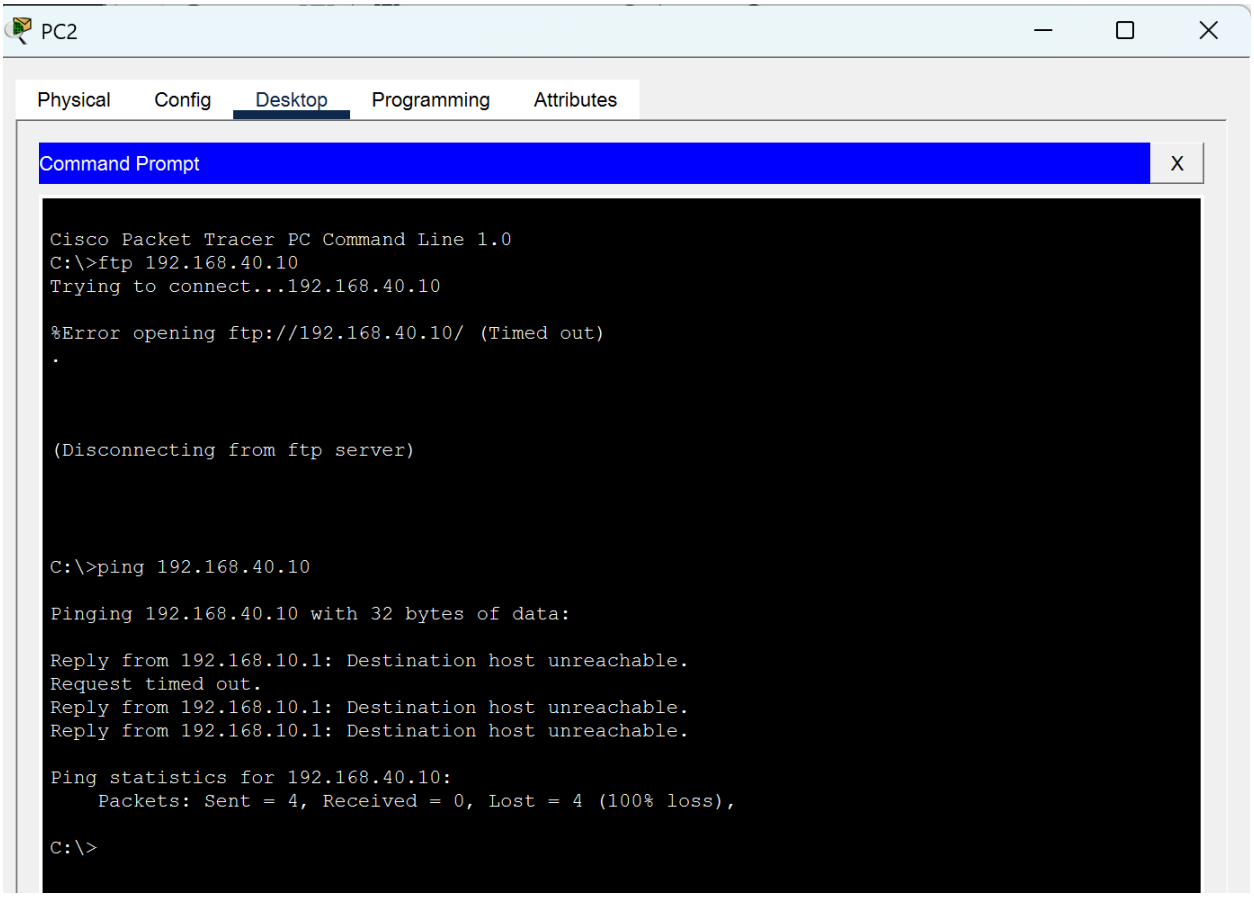




2. PC2 – FTP Access Test

- PC2 attempted to access the **FTP Server**.
- FTP access on **port 21** was **successfully blocked**.
- Other services from PC2 were not affected.

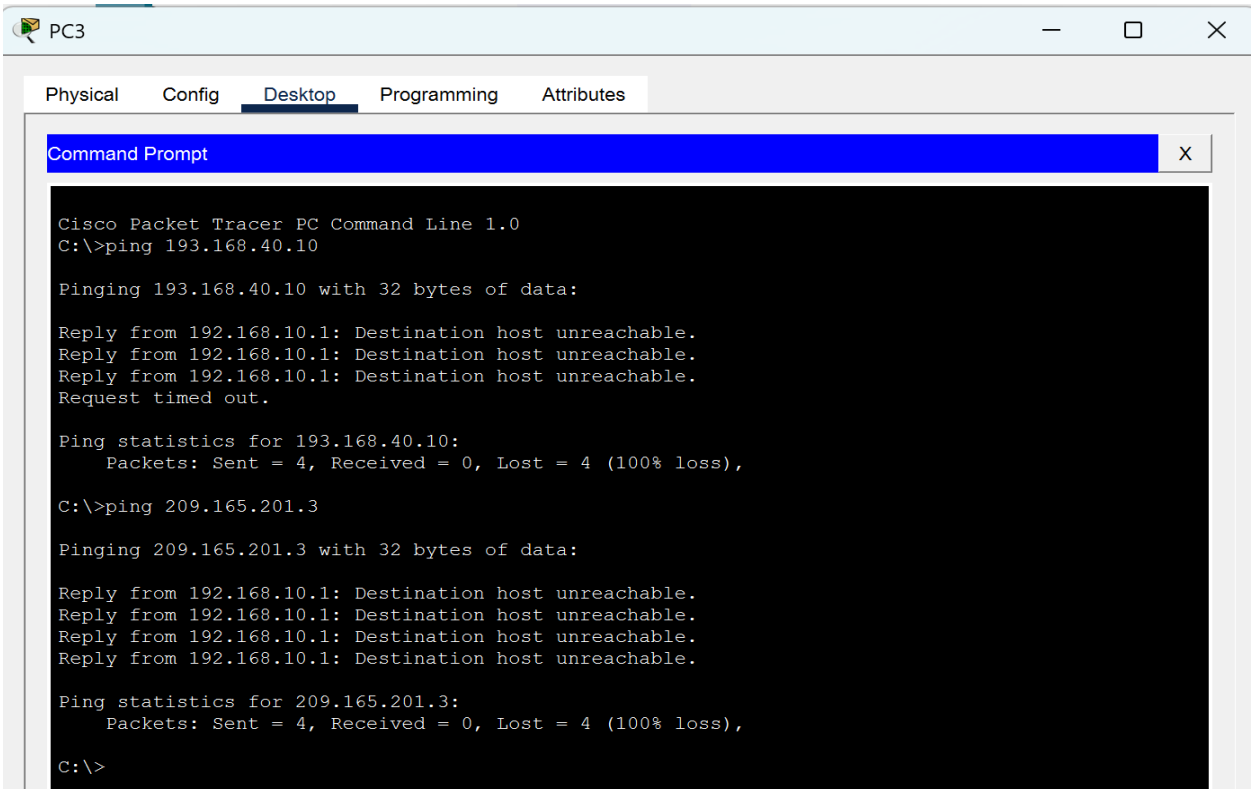
This confirms correct protocol-specific filtering.



3. PC3 – ICMP (Ping) Test

- PC3 attempted to ping **miu.edu.eg** and the **FTP Server**.
- Both ping attempts **failed**, indicating that ICMP traffic was correctly denied.
- Other PCs were still able to ping these destinations.

This confirms correct ICMP filtering without affecting other hosts.



4. ACL Hit Count Verification

The show access-lists command on R1 showed **incrementing hit counters** on the deny statements corresponding to:

- PC1 HTTP/HTTPS traffic
- PC2 FTP traffic
- PC3 ICMP traffic

This confirms that the ACL is **actively filtering live traffic** and functioning as intended.

```
R1>en
R1#show access-lists
Extended IP access list 110
    10 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Extended IP access list BLOCK_TRAFFIC
    10 deny tcp host 192.168.10.10 host 209.165.201.3 eq www
    20 deny tcp host 192.168.10.10 eq www host 209.165.201.3
    30 deny tcp host 192.168.10.10 host 209.165.201.3 eq 443
    40 deny tcp host 192.168.10.10 eq 443 host 209.165.201.3
    50 deny tcp host 192.168.10.11 host 209.165.201.4 eq ftp
    60 deny icmp host 192.168.10.12 host 209.165.201.3 (4 match(es))
    70 deny icmp host 192.168.10.12 host 209.165.201.4
    80 permit ip any any (281 match(es))

R1#
```

Part 9: Configure ASA Basic Settings and Firewall Using the CLI

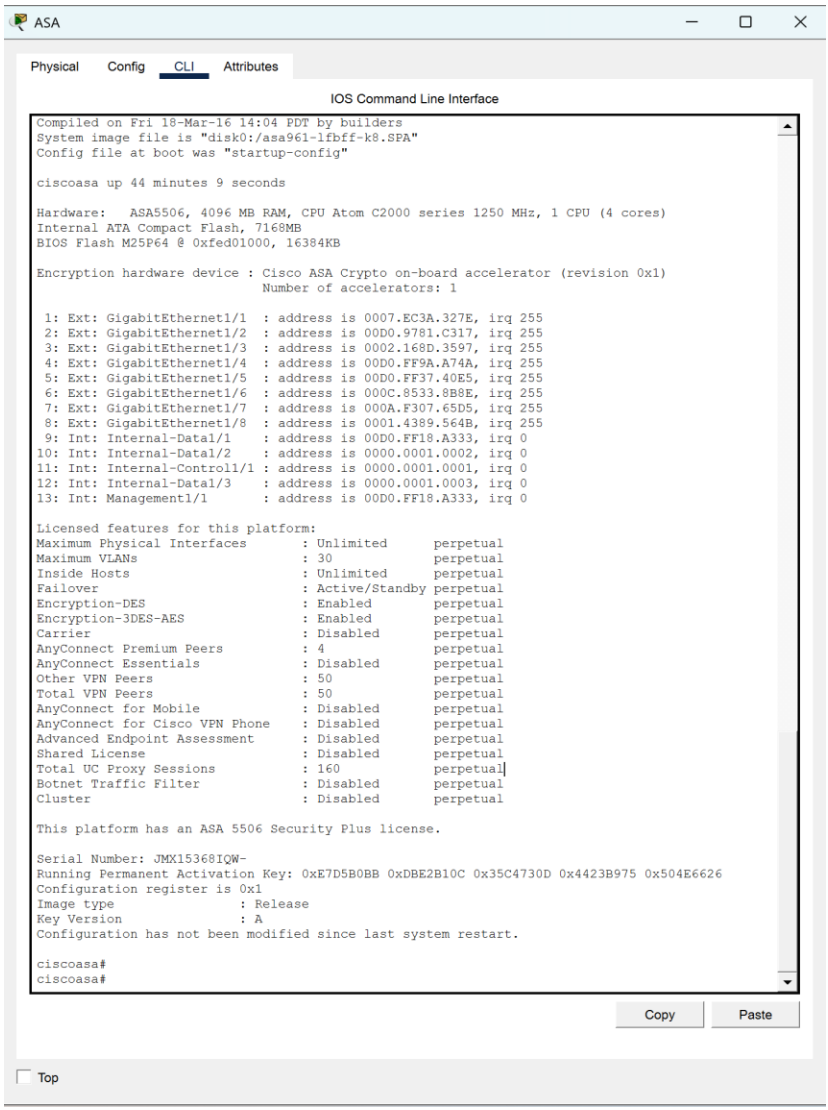
Overview

In this part, a Cisco Adaptive Security Appliance (ASA) was prepared with basic system settings and initial firewall configuration using the command-line interface. The objective was to identify the ASA platform details, configure essential system parameters, define interface roles, and verify basic connectivity while enforcing security principles between the inside and outside networks.

Step 1: Determining ASA Version, Interfaces, and License

The ASA device information was examined to identify the software version, hardware capabilities, available interfaces, and installed licenses. This step ensured that the ASA supports the required firewall and routing features before proceeding with further configuration. Verifying the version and license is essential to confirm compatibility with security services such as stateful inspection and interface security levels.





Step 2: Verifying File System and Flash Memory

The ASA was assigned a hostname and domain name to uniquely identify it within the network infrastructure. Setting the hostname improves device management and troubleshooting, while the domain name is required for features such as cryptographic services and certificate-based operations.

```
ciscoasa#
ciscoasa#show file system

File Systems:

       Size(b)       Free(b)      Type  Flags  Prefixes
*    128573440      42116608      disk  rw      disk0: flash:

ciscoasa#show flash:
--#--  --length--  -----date/time------  path
    1  86456832                         asa961-lfbff-k8.SPA
```

Step 4: Configuring Date and Time

The system clock was manually set on the ASA. Maintaining accurate date and time is important for logging, security auditing, and troubleshooting events within the firewall.



Step 5: Configuring INSIDE and OUTSIDE Interfaces

Two ASA interfaces were configured to establish security boundaries:

- The **OUTSIDE interface** was assigned to the external network and configured with the lowest security level (0), representing an untrusted network.
- The **INSIDE interface** was assigned to the internal network and configured with the highest security level (100), representing a trusted network.

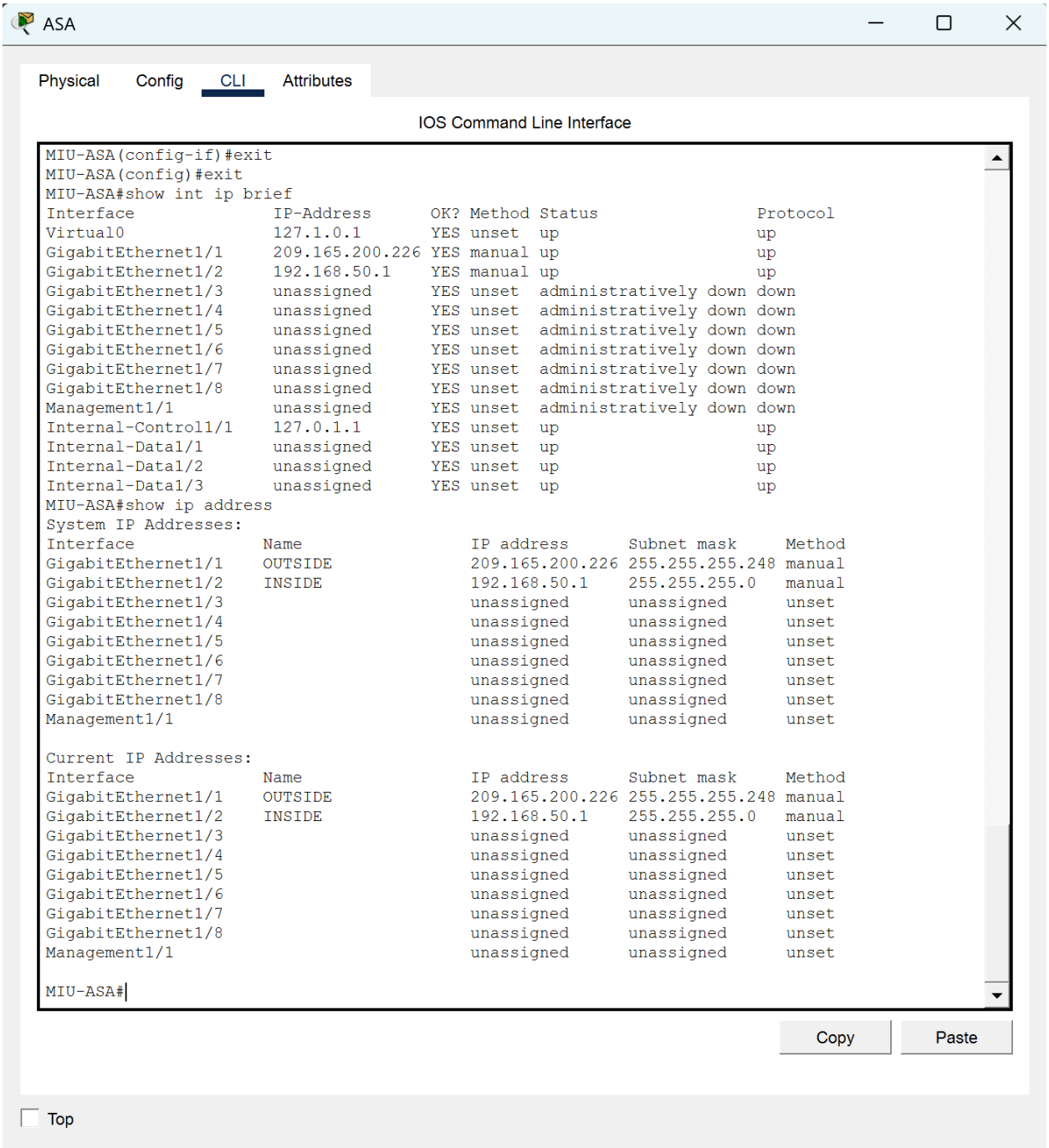
Security levels were used to enforce the default ASA behavior, which allows traffic from higher security zones to lower security zones while restricting traffic in the opposite direction unless explicitly permitted.

```
MIU-ASA(config)#int g1/1
MIU-ASA(config-if)#nameif OUTSIDE
INFO: Security level for "OUTSIDE" set to 0 by default.
MIU-ASA(config-if)#security-level 0
MIU-ASA(config-if)#ip address 209.165.200.226 255.255.255.248
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
MIU-ASA(config-if)#no shutdown

MIU-ASA(config-if)#exit
MIU-ASA(config)#int g1/2
MIU-ASA(config-if)#nameif INSIDE
INFO: Security level for "INSIDE" set to 0 by default.
MIU-ASA(config-if)#security-level 100
MIU-ASA(config-if)#ip address 192.168.50.1 255.255.255.0
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
MIU-ASA(config-if)#no shutdown
```

Step 6: Verifying Interface Configuration

The interface status and IP addressing were verified to ensure that both the INSIDE and OUTSIDE interfaces were operational and correctly configured. Confirming interface status at this stage ensured that the ASA was ready to process traffic according to its security policies.

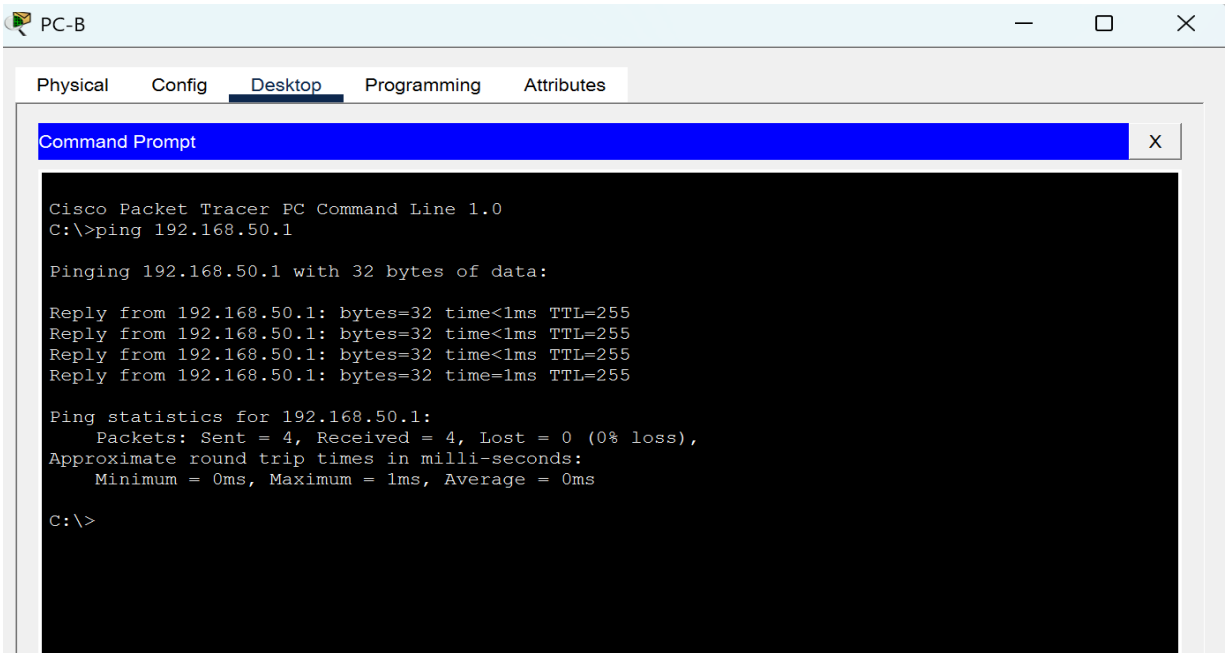


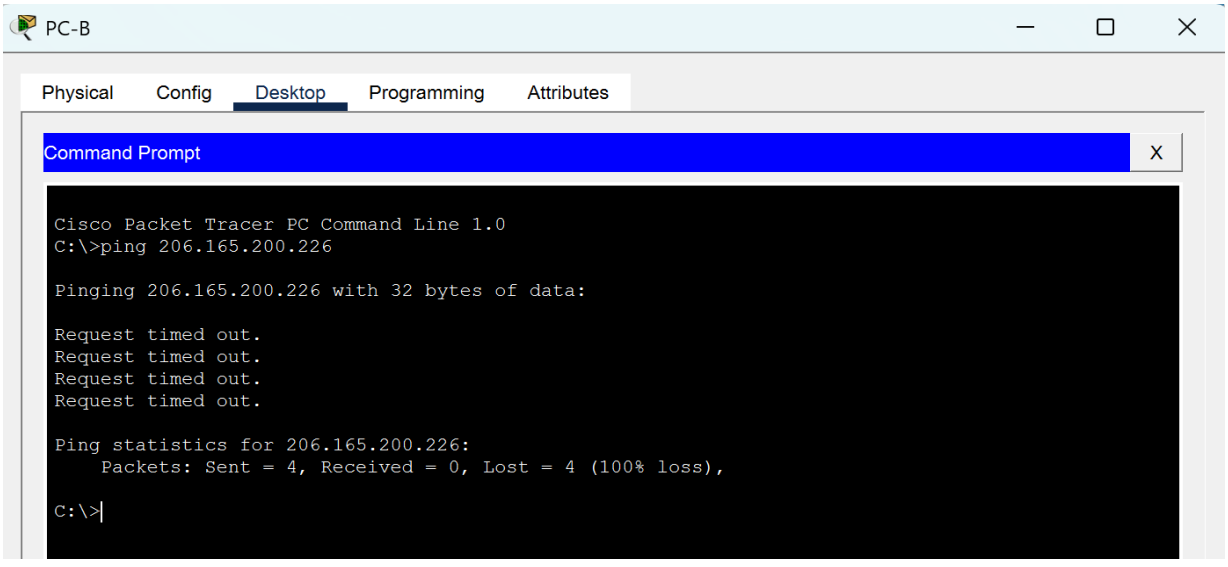
Step 7: Testing Connectivity to the ASA

Connectivity tests were performed to validate the firewall behavior:

- A host on the internal network successfully reached the ASA INSIDE interface, confirming proper internal connectivity.
- Attempts to reach the ASA OUTSIDE interface from the internal network were unsuccessful, demonstrating that the ASA correctly restricts access to lower-security interfaces.

These results confirmed that the ASA was enforcing security boundaries as intended.





Step 8: Modifying the Default MPF Inspection Policy

The Cisco ASA Modular Policy Framework (MPF) was examined and modified to allow ICMP inspection.

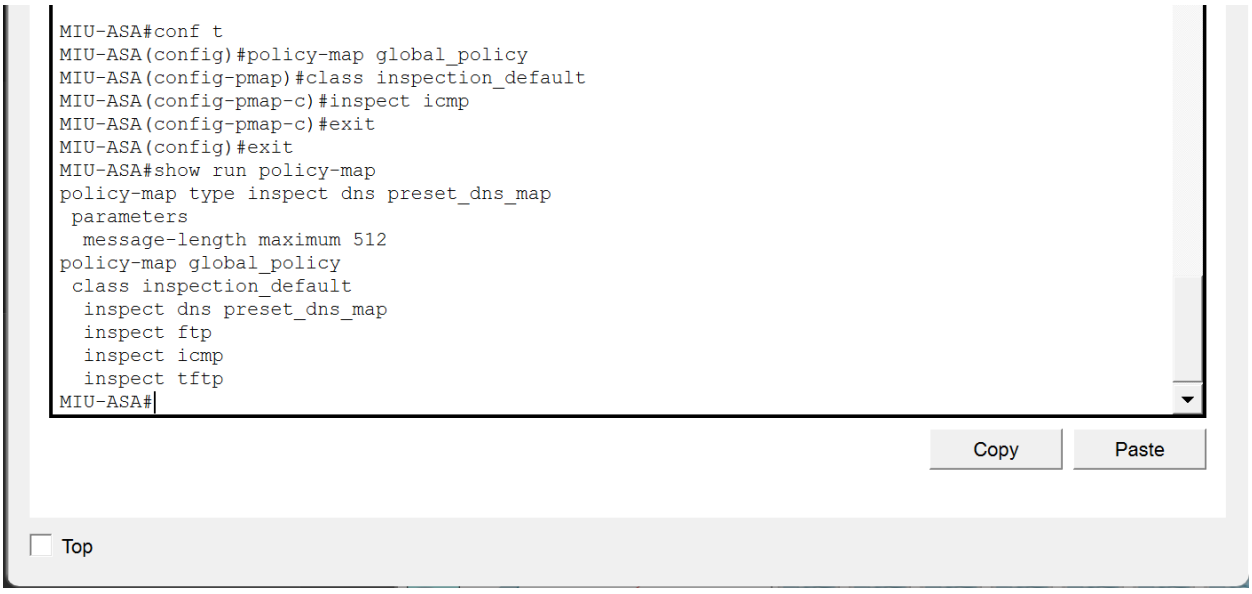
Initially, the default global policy (global_policy) was reviewed. The inspection class inspection_default was found to include inspections for DNS, FTP, and TFTP traffic. However, ICMP inspection was missing. This meant that although inside hosts could send ICMP echo requests, return echo replies from the outside interface were not permitted by the firewall.

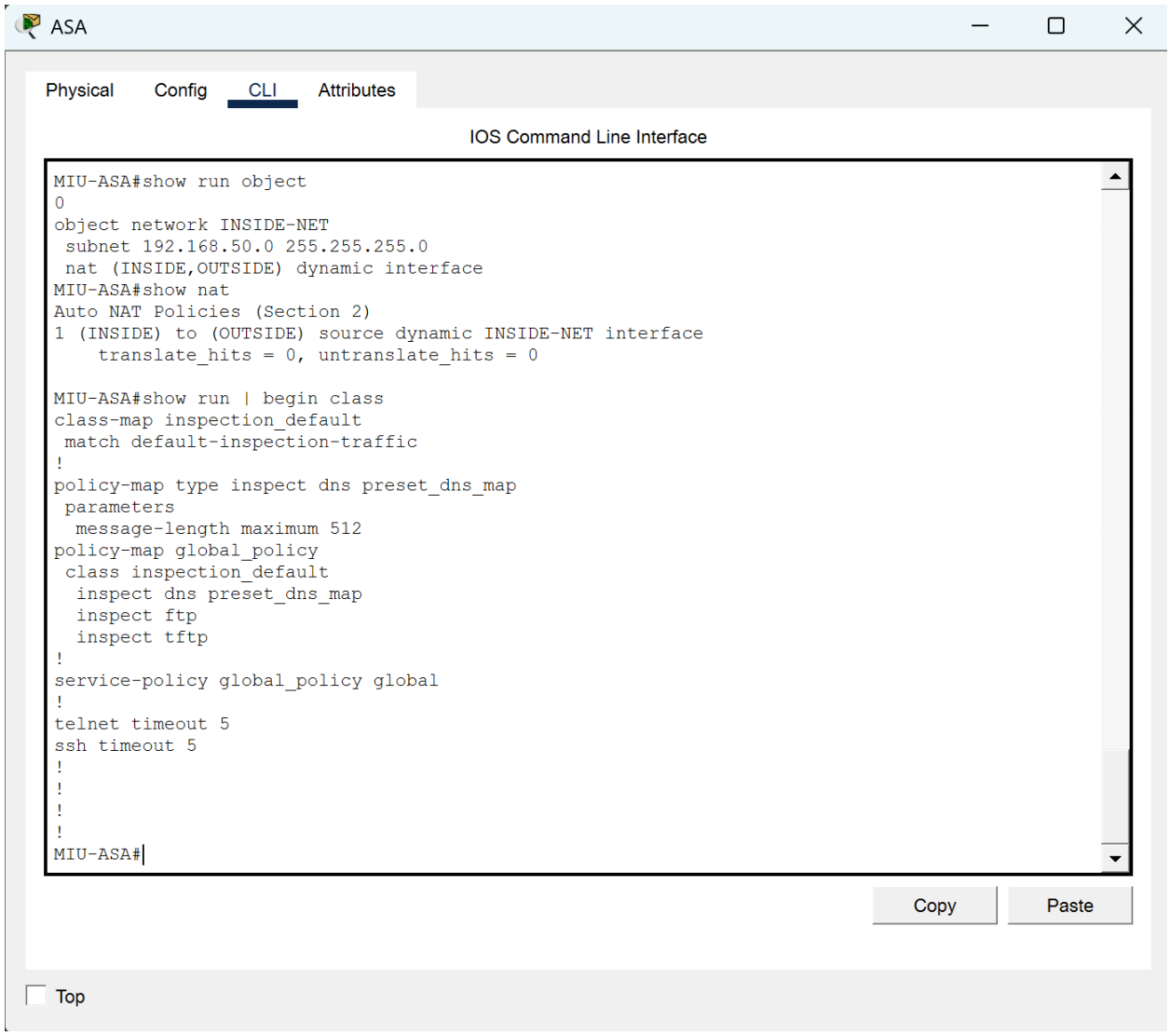
ICMP inspection was then added to the inspection_default class within the global policy. After this modification, the policy map was displayed again to confirm that **ICMP inspection was successfully enabled** and listed alongside the other inspected protocols.

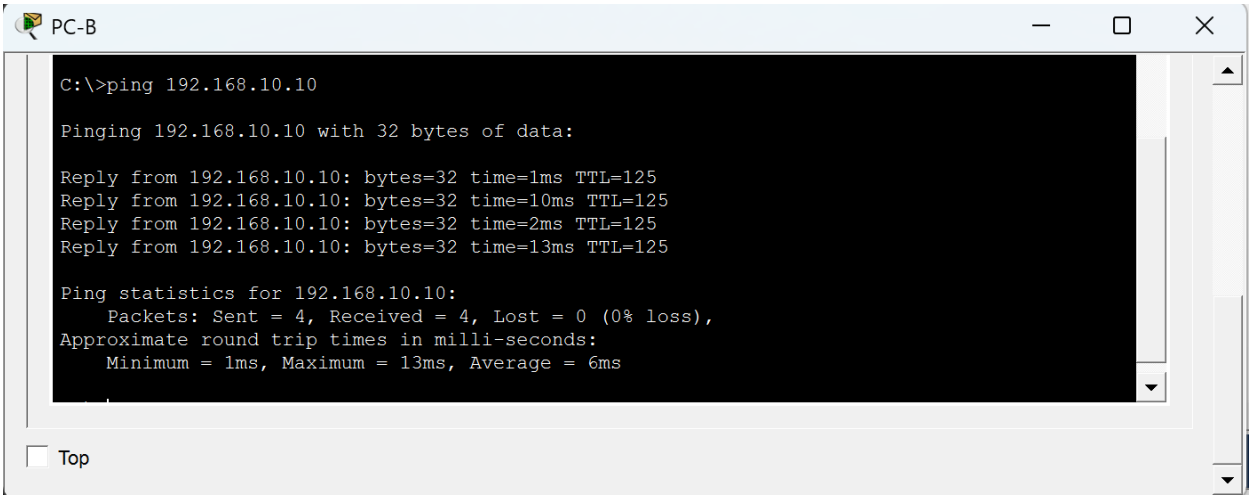
Verification

After enabling ICMP inspection:

- The policy map output confirmed that **inspect icmp** was present.
- Pings initiated from the inside network (PC-B) to internal hosts (PC1) were successful.
- This confirmed that **stateful inspection was working correctly**, allowing legitimate return ICMP traffic through the ASA.




ASA
Physical
Config
CLI
Attributes
IOS Command Line Interface
MIU-ASA#show run object
0
object network INSIDE-NET
subnet 192.168.50.0 255.255.255.0
nat (INSIDE,OUTSIDE) dynamic interface
MIU-ASA#show nat
Auto NAT Policies (Section 2)
1 (INSIDE) to (OUTSIDE) source dynamic INSIDE-NET interface
translate_hits = 0, untranslate_hits = 0
MIU-ASA#show run | begin class
class-map inspection_default
match default-inspection-traffic
!
policy-map type inspect dns preset_dns_map
parameters
message-length maximum 512
policy-map global_policy
class inspection_default
inspect dns preset_dns_map
inspect ftp
inspect tftp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
!
!
!
MIU-ASA#
Copy
Paste
Top


PC-B
C:\>ping 192.168.10.10
Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time=1ms TTL=125
Reply from 192.168.10.10: bytes=32 time=10ms TTL=125
Reply from 192.168.10.10: bytes=32 time=2ms TTL=125
Reply from 192.168.10.10: bytes=32 time=13ms TTL=125
Ping statistics for 192.168.10.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 13ms, Average = 6ms
Top

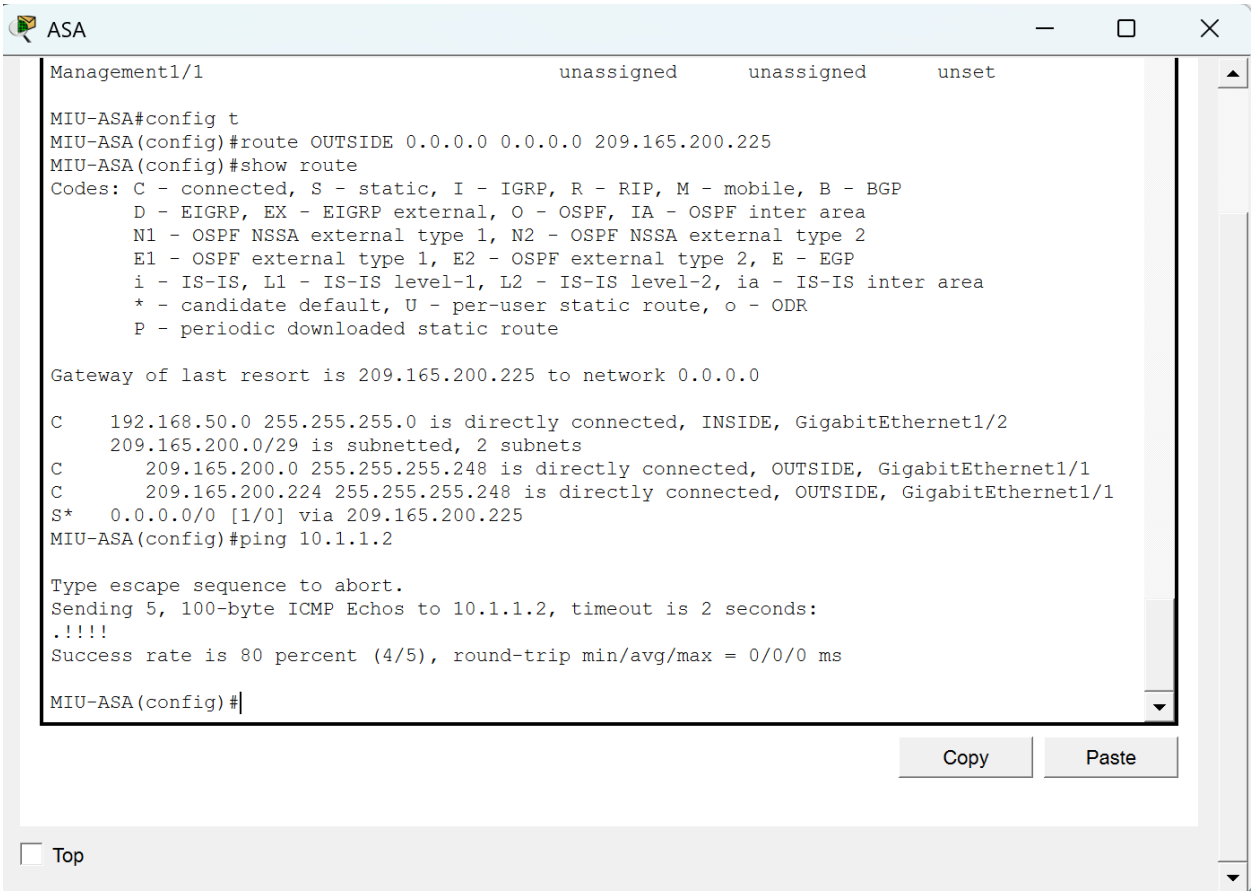
Step 9: Configure Routing, Address Translation, and Inspection Policy

Static Default Route Configuration

A static default route (quad-zero route) was configured on the ASA OUTSIDE interface to allow traffic destined for external networks to be forwarded to the ISP gateway.

Verification

- The routing table was displayed and showed the default route pointing to the ISP gateway.
- The ASA was able to successfully ping the ISP router interface, confirming correct routing.



Step 10: PAT Configuration Using Network Objects

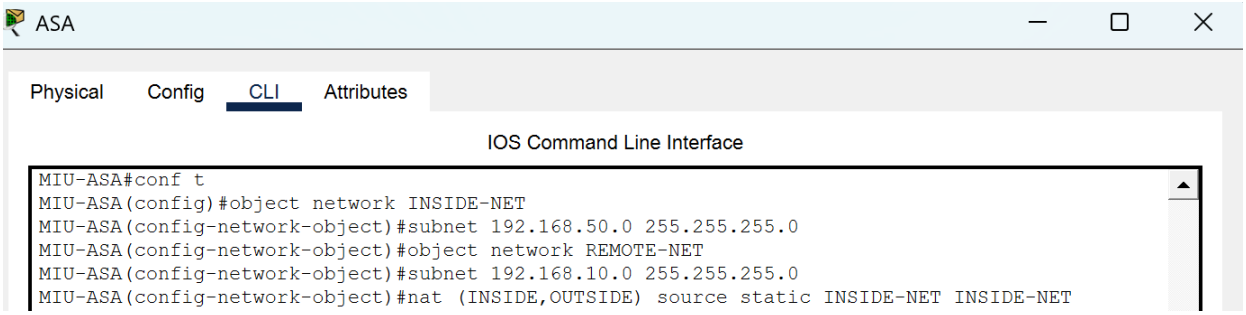
Port Address Translation (PAT) was configured using a network object representing the inside network. This allowed inside hosts to share the ASA's outside interface IP address for outbound traffic.

The NAT configuration was verified by examining the running configuration, where the network object and NAT rules were correctly applied.

Verification

- Pings from PC-B to the ISP interface were initiated.
- The pings failed as expected due to firewall policy restrictions.
- The show nat output confirmed that:
    - Outgoing ICMP echo requests were translated.
    - Return ICMP echo replies were blocked by the firewall before ICMP inspection was enabled.

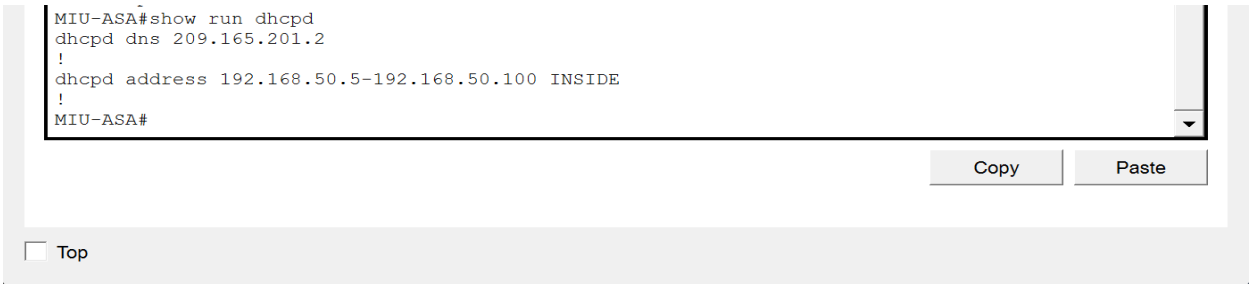
This behavior confirmed that **NAT translation was functioning correctly** and that traffic handling matched the configured security policy.



Step 11: DHCP Server Configuration

The ASA was configured as a DHCP server on the INSIDE interface, providing IP addresses within a defined range. Internal clients successfully received dynamic IP configuration, confirming proper DHCP operation.

```
MIU-ASA#conf t
MIU-ASA(config)#dhcpd address 192.168.50.5-192.168.50.100 INSIDE
MIU-ASA(config)#dhcpd dns 209.165.201.2
MIU-ASA(config)#dhcpd enable INSIDE
```



Step 12: AAA Authentication

Local user authentication was configured using the ASA's internal database. This database was used for SSH authentication.

SSH Remote Access

SSH access was enabled for:

- Internal network hosts
- A specific external management host

Secure SSH sessions were successfully established from both inside and outside networks, confirming proper access control and authentication.

```
MIU-ASA>en
Password:
MIU-ASA#conf t
MIU-ASA(config)#username admin password miu_admin
MIU-ASA(config)#aaa authentication ssh console LOCAL
MIU-ASA(config)#exit
```

```
MIU-ASA#conf t
MIU-ASA(config)#crypto key generate rsa modulus 2048
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

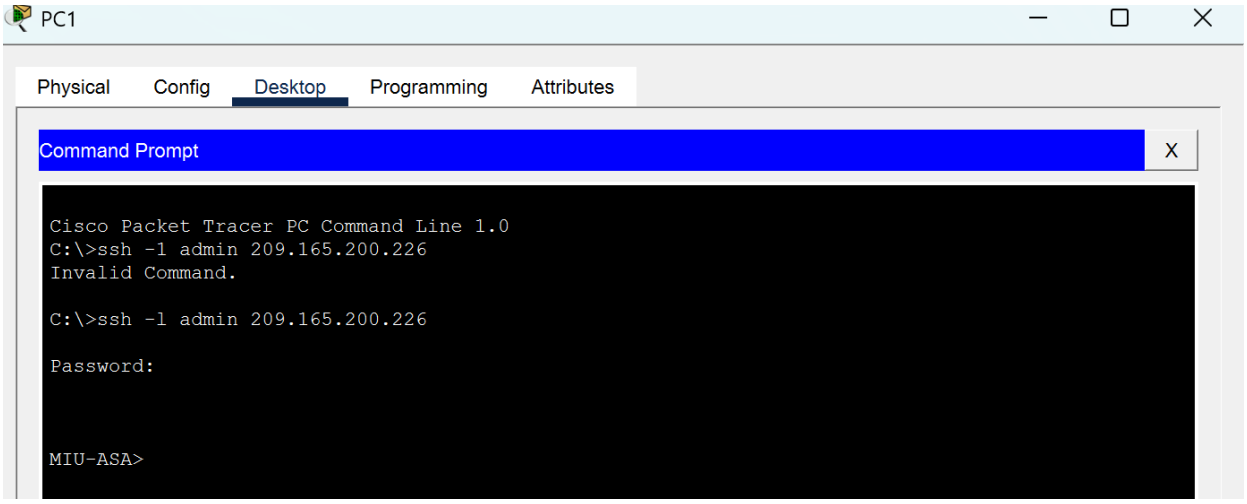
Do you really want to replace them? [yes/no]:
```

```
MIU-ASA#conf t
MIU-ASA(config)#crypto key generate rsa modulus 2048
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.

Do you really want to replace them? [yes/no]: yes
Keypair generation process begin. Please wait...
MIU-ASA(config)#ssh 192.168.50.0 255.255.255.0 INSIDE
MIU-ASA(config)#ssh 192.168.10.10 255.255.255.255 OUTSIDE
MIU-ASA(config)#ssh timeout 10
MIU-ASA(config)#exit
MIU-ASA#show ssh
Timeout: 10 minutes
Versions allowed: 1 and 2
MIU-ASA#
```

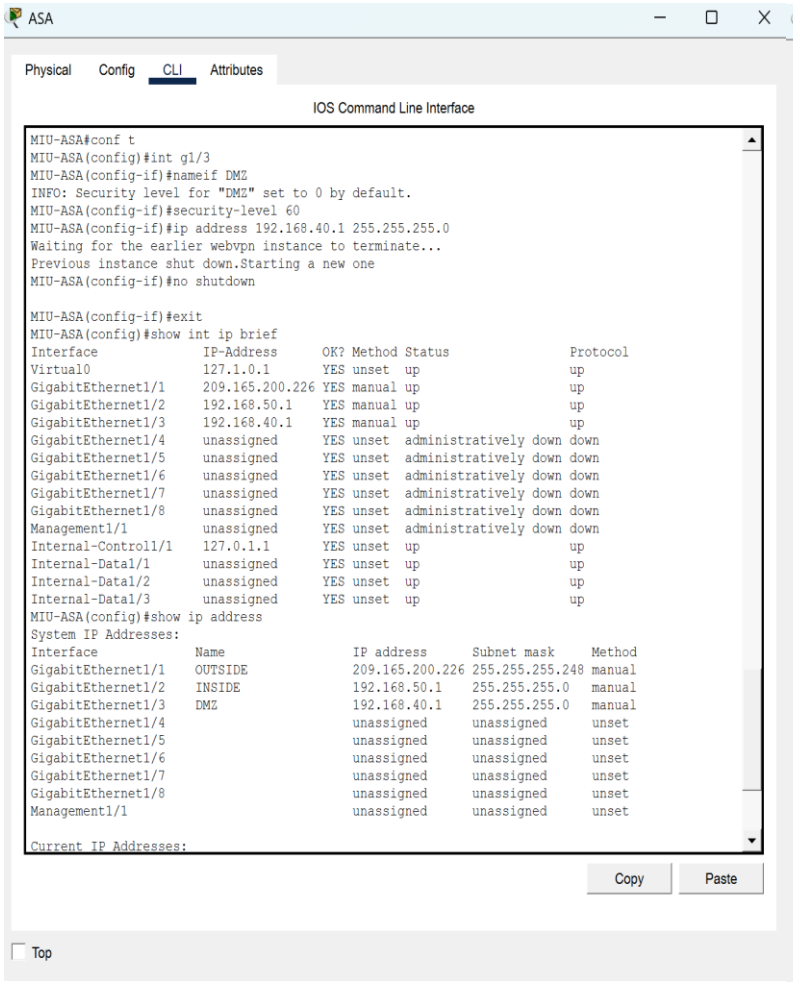
Copy Paste

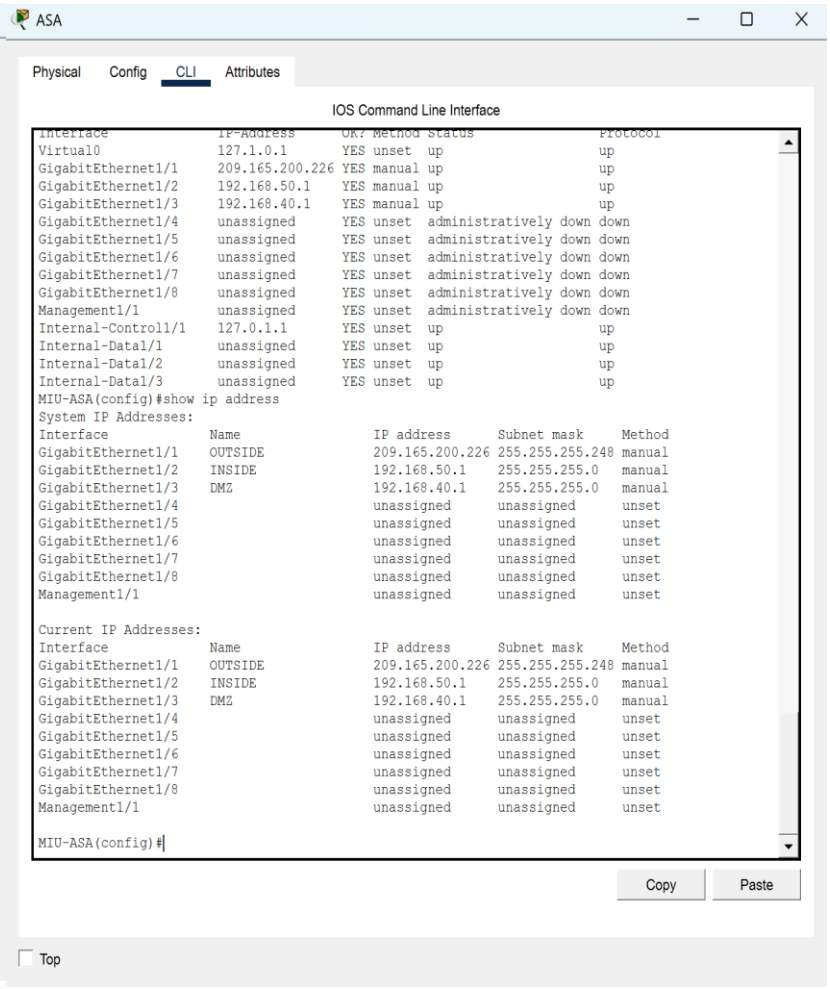
Top



Step 13: DMZ Interface Configuration

A DMZ interface was configured with a medium security level to host a public-facing web server. Interface status and addressing were verified.





Step 14: Static NAT for DMZ Server

Static NAT was configured to map the internal DMZ server address to a public IP address, allowing Internet access to the server.

```
MIU-ASA(config)#object network DMZ-SERVER
MIU-ASA(config-network-object)#host 192.168.40.10
MIU-ASA(config-network-object)#nat (DMZ,OUTSIDE) static 209.165.200.227
MIU-ASA(config-network-object)#exit
MIU-ASA#show run object network DMZ-SERVER
```

```
MIU-ASA#show nat
Auto NAT Policies (Section 2)
1 (DMZ) to (OUTSIDE) source static DMZ-SERVER 209.165.200.227
    translate_hits = 0, untranslate_hits = 0
2 (INSIDE) to (OUTSIDE) source dynamic INSIDE-NET interface
    translate_hits = 0, untranslate_hits = 0

MIU-ASA#
```

Step 15: DMZ Access Control List

An access control list was applied to the OUTSIDE interface to permit:

- HTTP traffic
- ICMP traffic
  from external hosts to the DMZ server.

This ensured controlled public access while maintaining firewall security.

```
MIU-ASA#conf t
MIU-ASA(config)#access-list OUTSIDE-DMZ extended permit tcp any host 192.168.40.10 eq 80
MIU-ASA(config)#access-list OUTSIDE-DMZ extended permit icmp any host 192.168.40.10
MIU-ASA(config)#access-group OUTSIDE-DMZ in interface OUTSIDE
MIU-ASA(config)#exit
MIU-ASA#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list OUTSIDE-DMZ; 2 elements; name hash: 0x23d64bd1
access-list OUTSIDE-DMZ line 1 extended permit tcp any host 192.168.40.10 eq www(hitcnt=0)
0xa63081ce
access-list OUTSIDE-DMZ line 2 extended permit icmp any host 192.168.40.10(hitcnt=0)
0x890ecd0a
```

Part 9 successfully demonstrated the configuration of a Cisco ASA firewall using the CLI. The ASA was secured with proper interface trust levels, routing, inspection policies, NAT, DHCP services, AAA authentication, SSH access, and DMZ protection. Verification steps confirmed that security policies were functioning as intended and that only authorized traffic was permitted.

Part 10: Configure Server-Based AAA Authentication Using RADIUS

Objective

The objective of this part was to configure **server-based AAA authentication using a RADIUS server** on router **R1**, while maintaining a **local database as a backup authentication method**. This ensures centralized user authentication with redundancy in case the RADIUS server becomes unavailable.

Step 1: Backup Local Authentication Configuration

A backup local user account was configured on **R1** to ensure continued access if the RADIUS server fails.
The local username **Admin1** was created with a secret password. This account is only used if RADIUS authentication is unavailable.

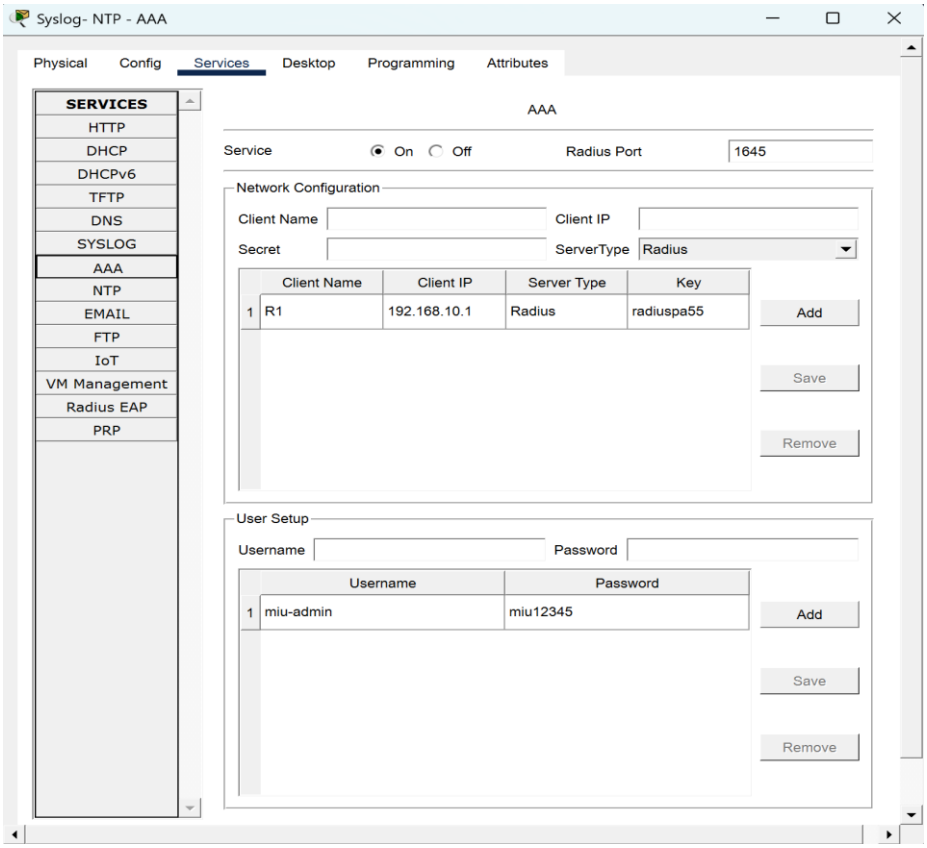
This step guarantees administrative access under failure conditions and aligns with best security practices.

Step 2: RADIUS Server Verification

The RADIUS server configuration was verified using the **Services > AAA** tab on the RADIUS server device.
Verification confirmed:

- A **network entry for R1**, allowing it to communicate with the RADIUS server.
- A **user account entry** configured for centralized authentication.

This confirmed that the RADIUS server was ready to authenticate users coming from R1.



Step 3: RADIUS Configuration on R1

Router **R1** was configured to use a RADIUS server named **Radius-ser**.
The RADIUS server IP address and shared secret key were defined to establish secure communication between R1 and the server.

Because Cisco Packet Tracer does not support the newer radius server command, the legacy RADIUS configuration commands were used as required by the simulator.

Note:
Although the lab instructions reference the radius server command, Cisco Packet Tracer does not support the new RADIUS server configuration syntax. Additionally, the legacy commands radius-server host and radius-server key are also unsupported.

Despite this limitation, AAA authentication behavior can still be verified logically by enabling AAA, defining an authentication method list that prioritizes RADIUS with local fallback, and validating user access using both RADIUS and local credentials.

```
R1(config)#aaa new-model
R1(config)#aaa authentication login default group radius local
R1(config)#line console 0
R1(config-line)#login authentication default
R1(config-line)#
```

```
R1#show run | section radius
aaa authentication login default group radius local
aaa authorization exec default group radius local
radius server 192.168.10.20
 address ipv4 192.168.10.20 auth-port 1645
radius server Radius-ser
 key radiuspa55
R1#
```

Part 10 successfully implemented **server-based AAA authentication using RADIUS** on R1 with a **local database fallback**.

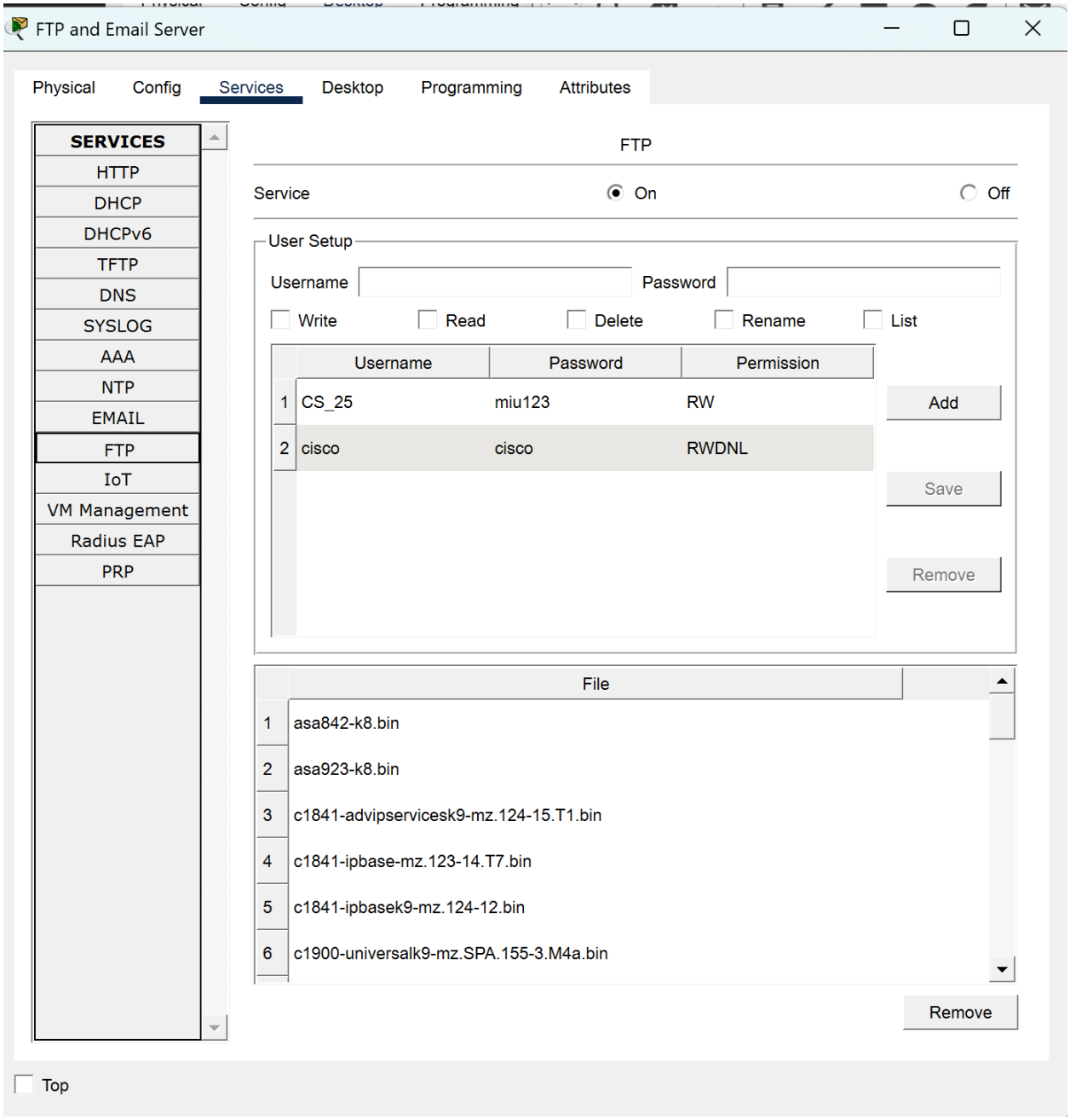
Part 11: Server Configuration

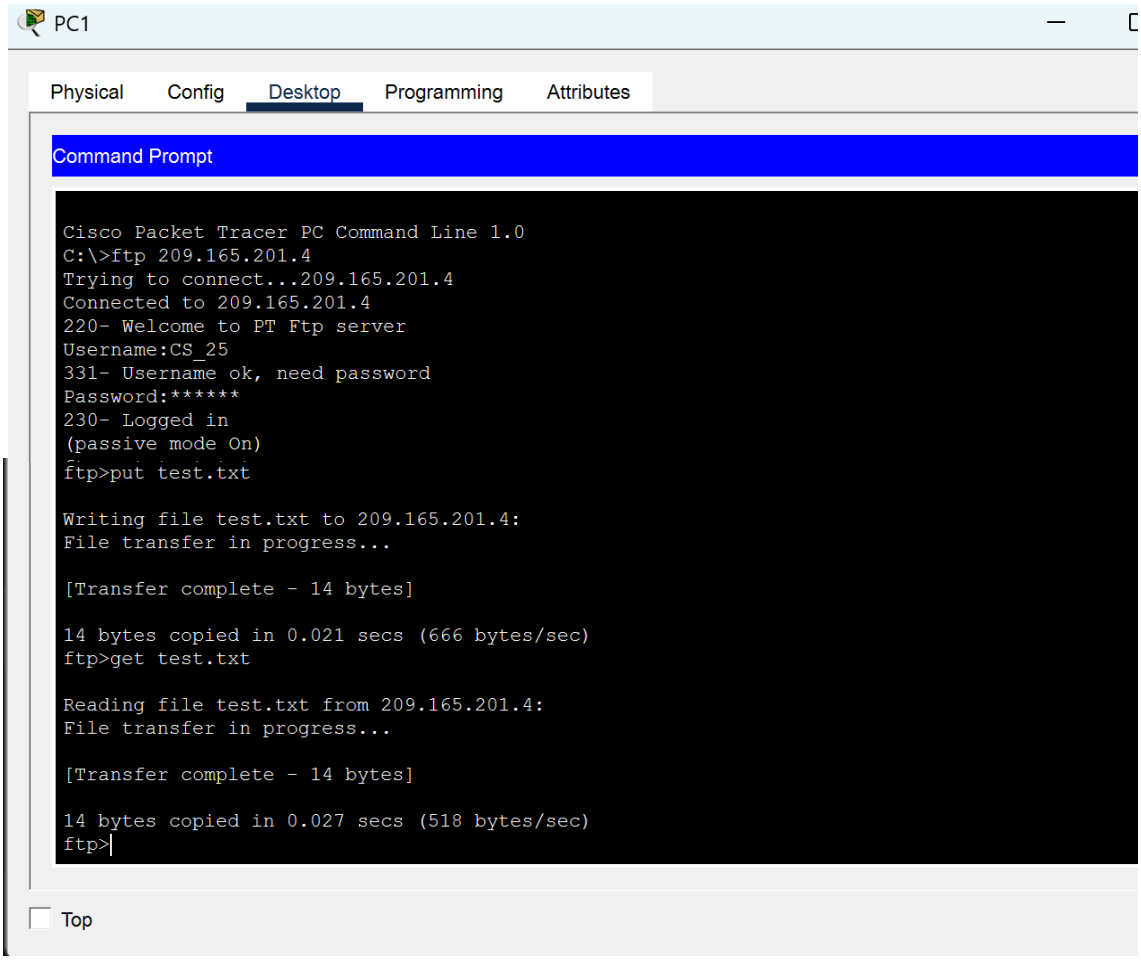
Step 1: FTP Server Configuration

In this part, the FTP server was configured to allow authenticated file transfer services. The FTP service was enabled on the server, and a new user account was created to control access. The credentials used were **username CS_25** with the password **miu123**.

After enabling the service and creating the user account, the FTP server status was checked to ensure that the service was running correctly. Connectivity was then tested by initiating an FTP session from a client device using the configured credentials. The

successful login confirmed that the FTP service was operating properly and that user authentication was functioning as expected.





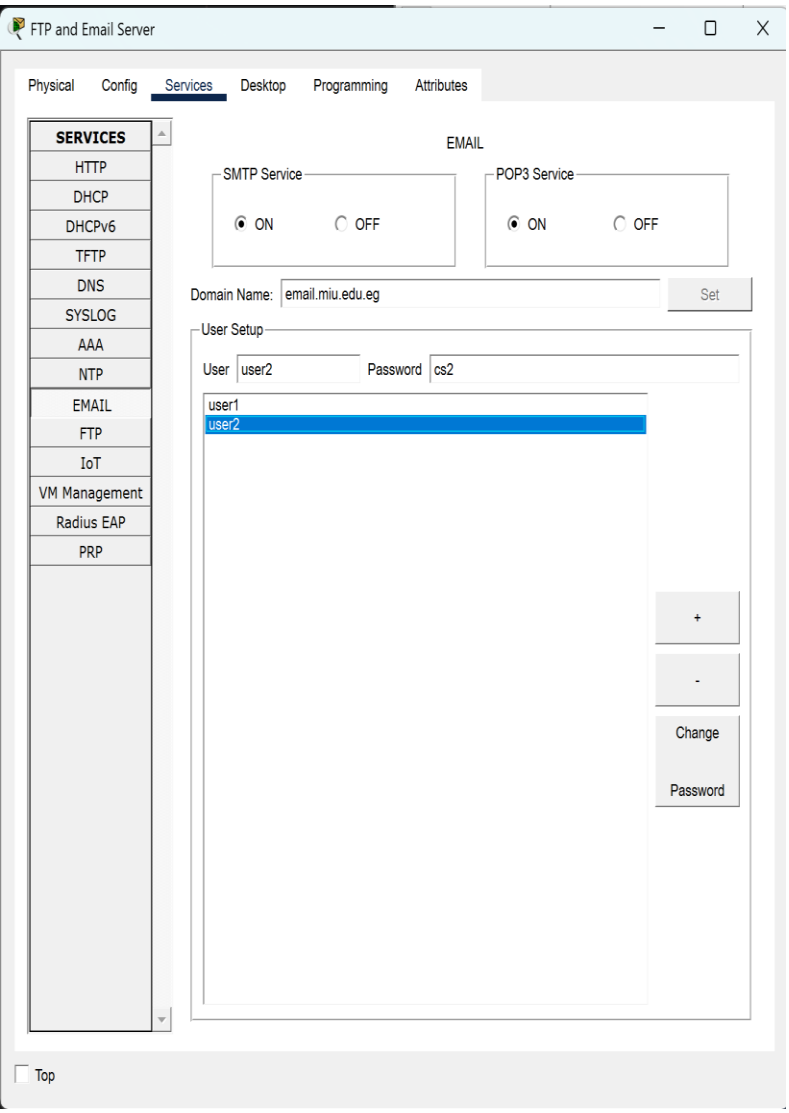
Step 2: Email Server Configuration

The email server was configured to support basic email communication using **SMTP** for sending emails and **POP3** for receiving them. Both services were enabled on the server, and the domain name **email.miu.edu.eg** was assigned.

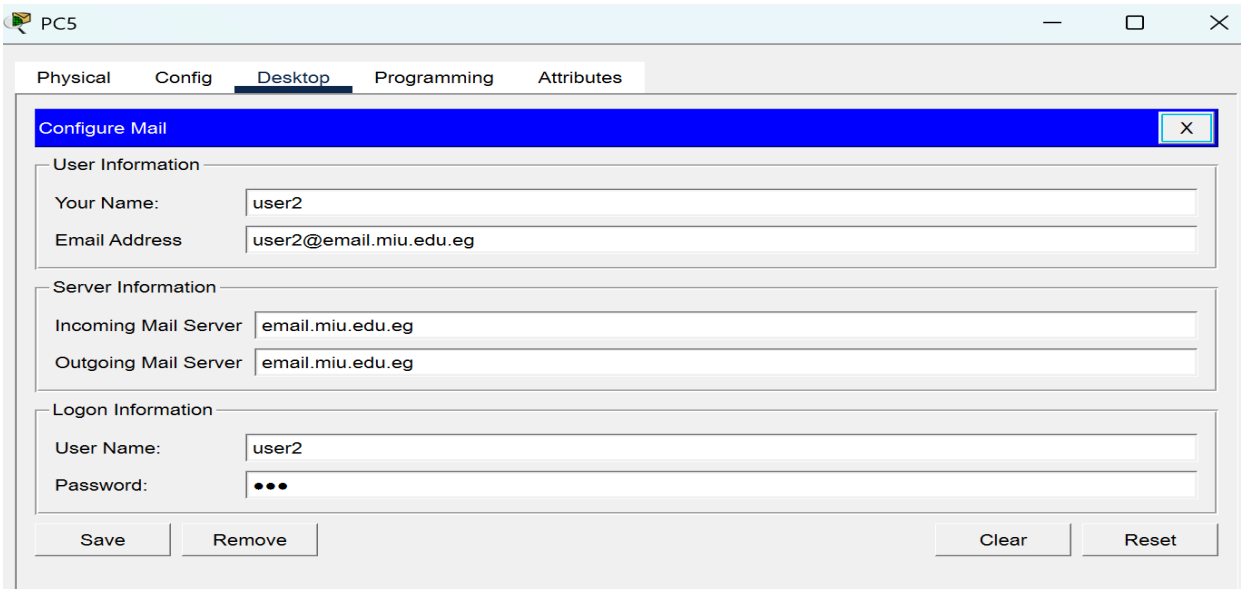
Two user accounts were created on the email server:

- **user1** with password **cs1**
- **user2** with password **cs2**

To verify the configuration, emails were sent between the two users using client email applications. Successful delivery and reception of messages confirmed that the SMTP and POP3 services were working correctly and that the email server was properly configured.


FTP and Email Server
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name:
email.miu.edu.eg
Set
User Setup
User
user2
Password
cs2
user1
user2
+
-
Change
Password
Top


PC1
Physical
Config
Desktop
Programming
Attributes
MAIL BROWSER
X
Mails
Compose
Reply
Receive
Delete
Configure Mail
From
Subject
Received
Sending mail to user2@email.miu.edu.eg , with subject : TEST .. Mail Server: email.miu.edu.eg
DNS resolving. Resolving name: email.miu.edu.eg by querying to DNS Server: 209.165.201.2
DNS resolved ip address: 209.165.201.4
Send Success.
Cancel
Send/Receive
Top


PC5
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name:
user2
Email Address
user2@email.miu.edu.eg
Server Information
Incoming Mail Server
email.miu.edu.eg
Outgoing Mail Server
email.miu.edu.eg
Logon Information
User Name:
user2
Password:
Save
Remove
Clear
Reset

Step 3: Web Server Configuration

The web server was configured to host the main website for the domain **miu.edu.eg**. The HTTP service was enabled, and the homepage was customized to display the **MIU logo**, ensuring that the website content loaded correctly.

After configuration, the website was accessed from a client device using a web browser and the domain name instead of the IP address. The successful display of the homepage confirmed that the web server was operational and accessible through domain-based addressing.

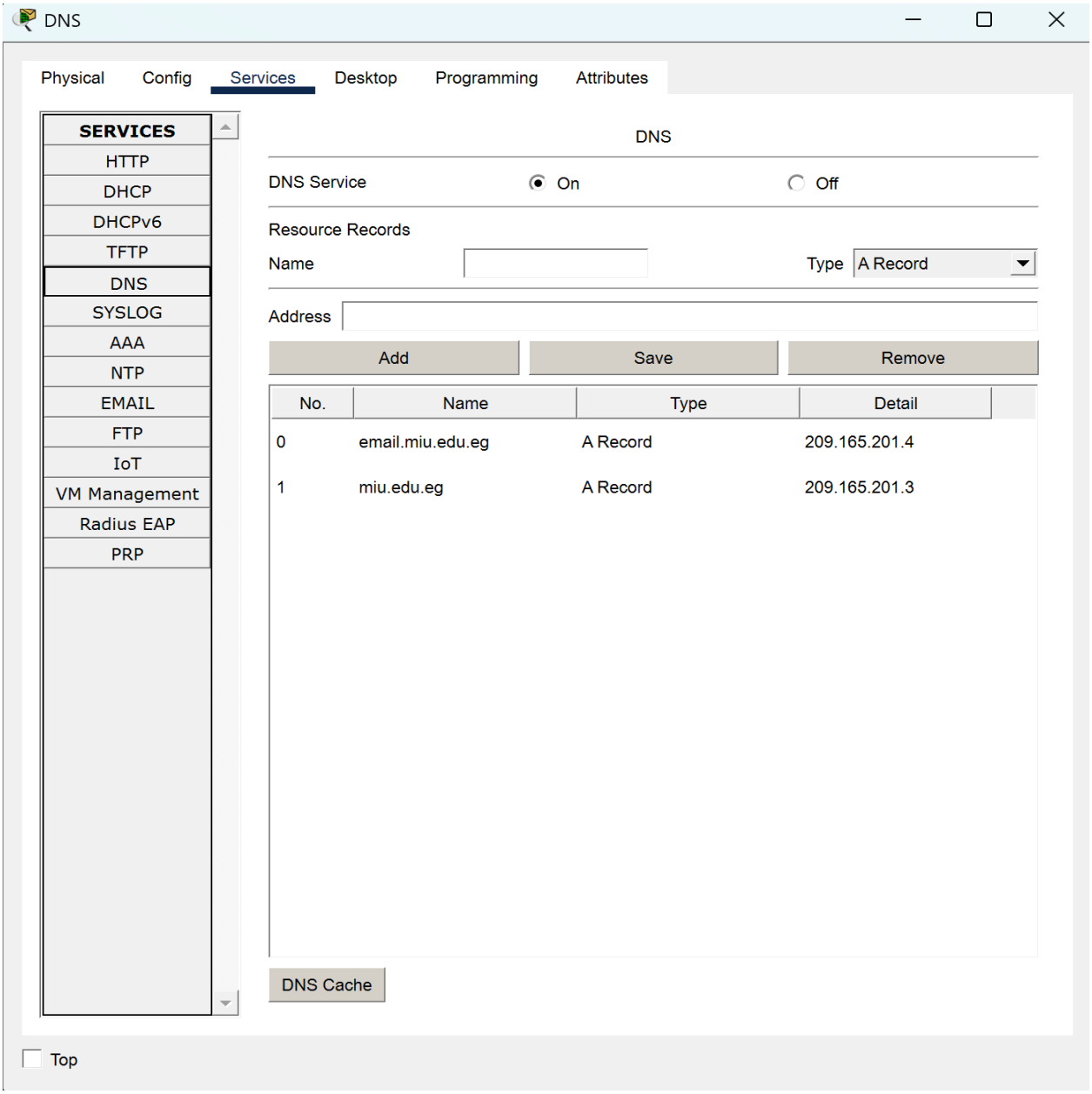


Step 4: DNS Server Configuration

The DNS server was configured to resolve domain names to their corresponding IP addresses. DNS records were created to map:

- **miu.edu.eg** to the web server IP address
- **email.miu.edu.eg** to the email server IP address

To verify DNS functionality, client machines were used to ping both domain names. The successful resolution of the domain names to the correct IP addresses confirmed that the DNS server was functioning properly and providing accurate name resolution services.



Part 12: Network Time and Logging Configuration

Step 1: Syslog Configuration

In this part, centralized logging was implemented using a Syslog server to monitor network events generated by intermediary devices.

First, the Syslog service was enabled on the designated server at Site 1. Once activated, the Syslog service was kept running to allow continuous monitoring of incoming log messages.

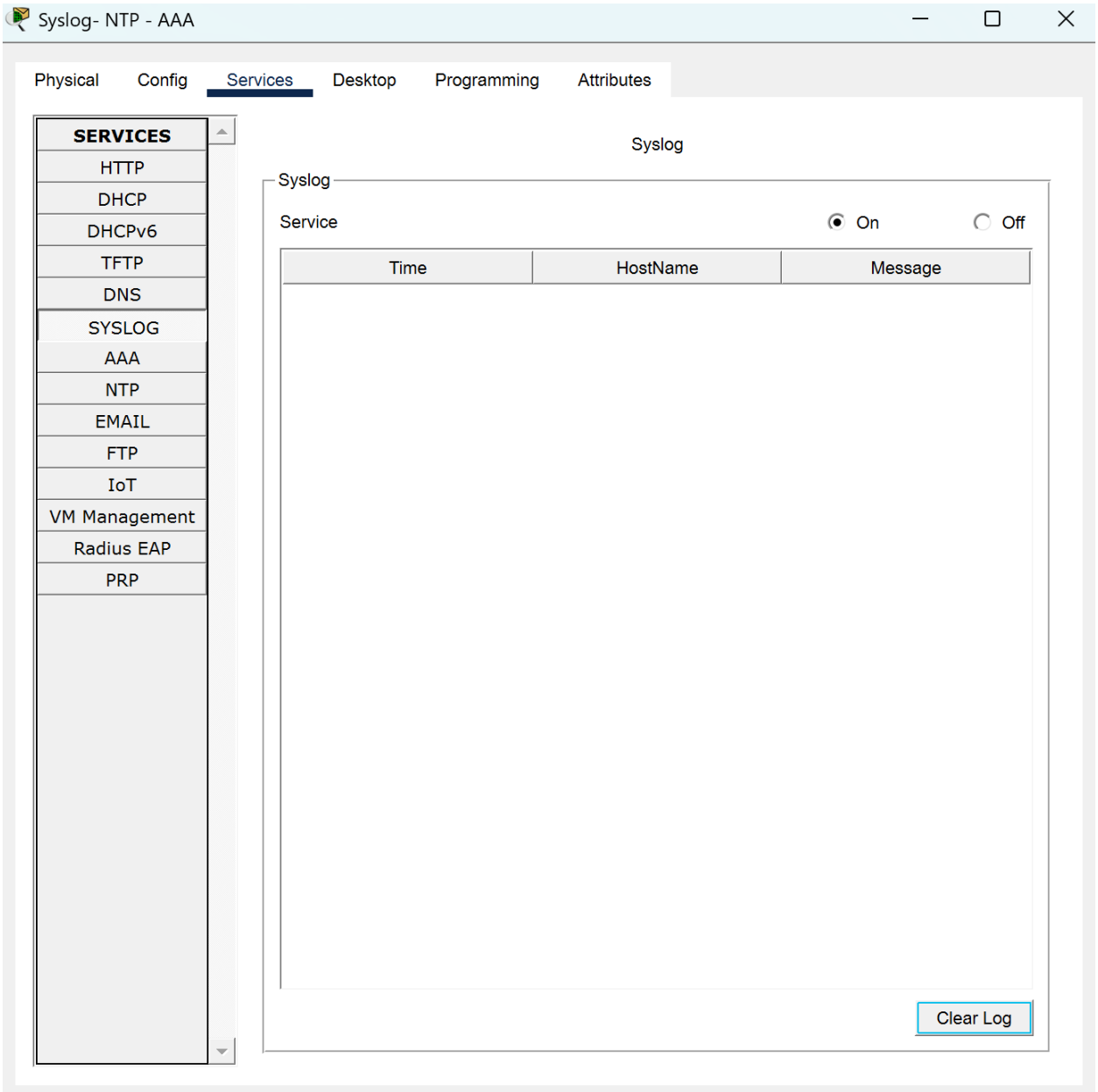
Next, the intermediary devices **R1** and **SW1** were configured to send their log events to the Syslog server. This ensured that important system messages, interface status changes, and operational events were centrally collected for analysis and troubleshooting purposes.

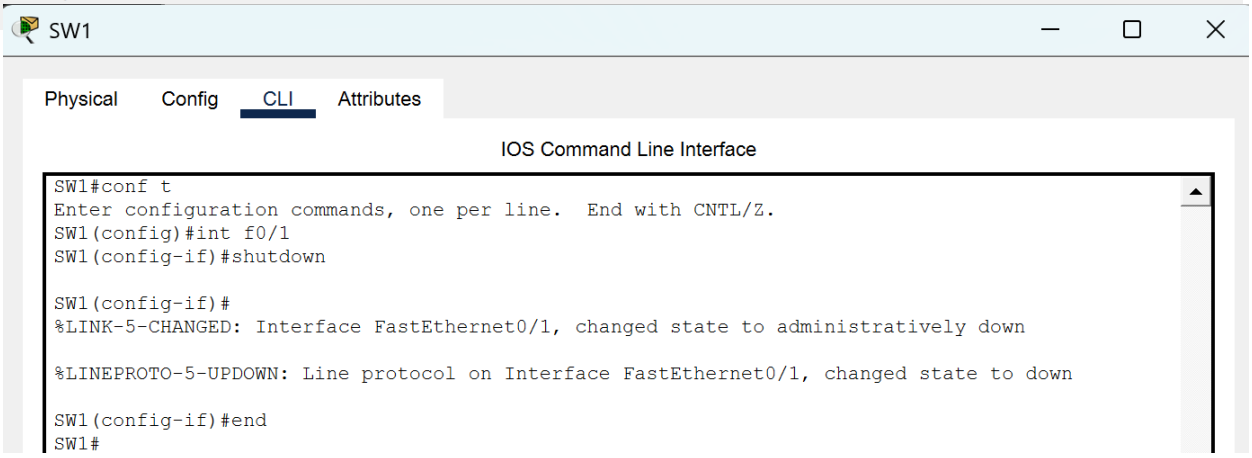
To generate Syslog events, interface state changes were performed. The **S0/1/0 interface on R1** and the **F0/1 interface on SW1** were administratively disabled. These actions successfully generated log entries, which were received and displayed on the Syslog server.

The Syslog events were examined on the server to verify that logs from both devices were being recorded correctly. Although all events were logged successfully, it was observed that the timestamps were incorrect at this stage due to the absence of proper time configuration. Finally, the Syslog log was cleared to prepare for the next part of the configuration.

```
R1(config)#logging host 192.168.10.20
```

```
R1(config)#logging on
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.10.20 port 514 started - CLI initiated
```


Syslog- NTP - AAA
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
Syslog
Syslog
Service
On
Off
Time
HostName
Message
Clear Log
Top


SW1
Physical
Config
CLI
Attributes
IOS Command Line Interface
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int f0/1
SW1(config-if)#shutdown
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
SW1(config-if)#end
SW1#

```
R1#en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int s0/1/0
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

21:24:41: %OSPF-5-ADJCHG: Process 10, Nbr 209.165.201.1 on Serial0/1/0 from FULL to DOWN,
Neighbor Down: Interface down or detached

R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Top

Syslog- NTP - AAA

Physical Config Services Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP
- PRP

Syslog

Syslog

Service On Off

	Time	HostName	Message
1	03.02.1993 09:29:36.904 PM	192.168.10.1	%SYS-5-CONFIG_I: Configured from console b…
2	12.13.2025 02:14:15.136 AM	192.168.10.2	%SYS-5-CONFIG_I: Configured from console b…
3	12.13.2025 02:14:33.230 AM	192.168.10.2	%LINK-5-CHANGED: …
4	12.13.2025 02:14:33.230 AM	192.168.10.2	%LINEPROTO-5-UPDOW…
5	12.13.2025 02:14:35.787 AM	192.168.10.2	%SYS-5-CONFIG_I: Configured from console b…
6	12.13.2025 02:20:38.032 AM	192.168.10.1	%SYS-5-CONFIG_I: Configured from console b…
7	12.13.2025 02:21:14.904 AM	192.168.10.1	%LINK-5-CHANGED: …
8	12.13.2025 02:21:19.843 AM	192.168.10.1	%LINK-5-CHANGED: …
9	12.13.2025 02:21:22.356 AM	192.168.10.1	%SYS-5-CONFIG_I: Configured from console b…

Clear Log

Top

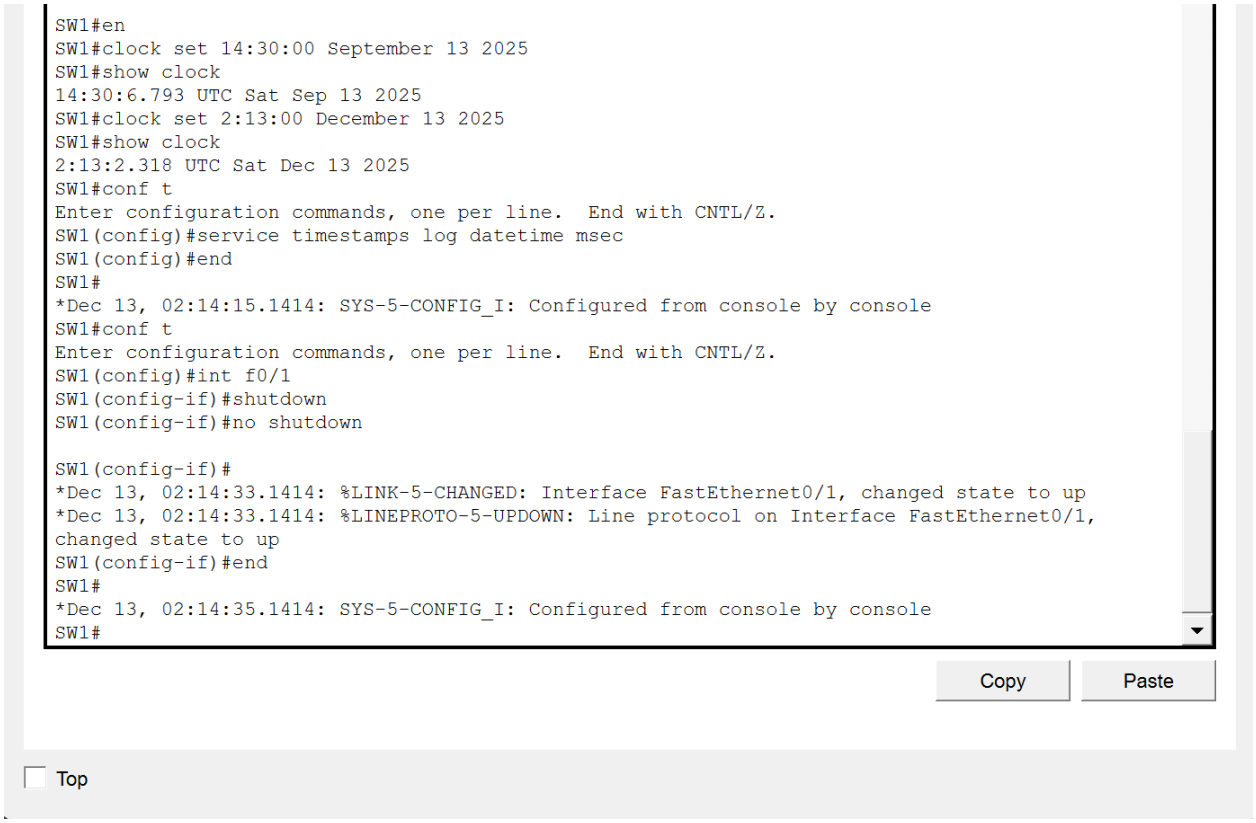
Step 2: Manual Clock Configuration and Timestamping

In this section, accurate time configuration was applied to improve log readability and reliability.

The system clock on **SW1** was manually set to the current date and approximate time. This step ensured that subsequent log messages generated by the switch contained meaningful timestamps.

After setting the clock, logging timestamp functionality was enabled on both **R1** and **SW1**. This configuration allowed each log message sent to the Syslog server to include a timestamp, making it easier to correlate events and analyze network activity over time.

With timestamps enabled, newly generated Syslog entries displayed correct and consistent time information, confirming that the network devices were properly synchronized for logging purposes.



Conclusion

In conclusion, this project successfully demonstrated the implementation of a complete enterprise-style network with multiple layers of security and management. All required configurations were completed step by step, starting from IP addressing and routing, and ending with logging, time configuration, and centralized monitoring.

Key security features such as SSH, OSPF authentication, site-to-site VPNs, zone-based firewalls, stateless ACLs, and ASA firewall policies were configured and verified. Server-based authentication using RADIUS highlighted the importance of centralized user management, while local authentication was kept as a backup to ensure reliability. In addition, server services such as FTP, Email, Web, and DNS were configured and tested to confirm correct functionality and accessibility.

Finally, Syslog and manual clock configuration allowed proper event tracking and log analysis, which are essential for troubleshooting and network auditing. Overall, this project reflects real-world networking practices and provides a strong foundation in network security, device configuration, and enterprise network management.